

Docker- Containerization

Assignment 43 – [Cloud VM Bootstrap + Docker Deploy \(Static Website on Port 80\)](#)

Step 1 — Launch the VM

Choose **one** provider:

AWS (EC2)

- AMI: Ubuntu Server 22.04 LTS
- Instance type: t2.micro/t3.micro (free-tier friendly)
- **Security Group:** allow **22/tcp** (your IP) and **80/tcp** (0.0.0.0/0)
- Paste the **User Data** script (below) before launching

Launch an instance [info](#)

[illegible]

Azure (VM)

- Image: Ubuntu 22.04 LTS
- Size: Standard_B1s/B2s
- **NSG:** inbound rules for **22/tcp** (your IP) and **80/tcp** (Any)
- Paste the **Custom Data** (cloud-init) script (below) at create time

Step 2 — Cloud-Init Script (User Data / Custom Data)

Use **one** of these. Both are cloud-init compatible.

```
#cloud-config
```

```
package_update: true
```

```
package_upgrade: true
```

```
packages:
```

- apt-transport-https
- ca-certificates
- curl
- gnupg
- lsb-release

```
runcmd:
```

```
# Install Docker (Ubuntu repo is fine for this lab)
```

- apt-get update -y
- apt-get install -y docker.io
- systemctl enable docker
- systemctl start docker
- usermod -aG docker ubuntu || true # change user if different
- newgrp docker <<EOF

```
echo "Docker ready"
```

```
EOF
```

Tip: On Azure, paste this under **Advanced** → **Custom data**. On AWS, paste in the **User data** box.

Step 3 — Verify Docker

SSH to the VM and verify:

```
docker --version
```

```
docker ps
```

```
ubuntu@ip-172-31-19-104:~$ docker --version
docker ps
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
ubuntu@ip-172-31-19-104:~$
```

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Step 4 — Get the App & Create a Dockerfile

On the VM

```
git clone https://github.com/pravinmishraaws/Azure-Static-Website.git
```

```
cd Azure-Static-Website
```

```
ubuntu@ip-172-31-19-104:~$ git clone https://github.com/pravinmishraaws/Azure-Static-Website.git
cd Azure-Static-Website
Cloning into 'Azure-Static-Website'...
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 10 (delta 2), reused 6 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (10/10), 5.60 KiB | 1.40 MiB/s, done.
Resolving deltas: 100% (2/2), done.
ubuntu@ip-172-31-19-104:~/Azure-Static-Website$
```

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Create Dockerfile (serves static site via Nginx)

```
cat > Dockerfile <<'EOF'
```

```
FROM nginx:alpine
```

```
# remove default nginx page
```



```
RUN rm -rf /usr/share/nginx/html/*
```

```
# copy repo contents to nginx web root
```

```
COPY . /usr/share/nginx/html
```

```
EXPOSE 80
```

```
# nginx default CMD is fine
```

```
EOF
```

```
ubuntu@ip-172-31-19-104:~/Azure-Static-Website$ # Create Dockerfile (serves static site via Nginx)
cat > Dockerfile <<'EOF'
FROM nginx:alpine
# remove default nginx page
RUN rm -rf /usr/share/nginx/html/*
# copy repo contents to nginx web root
COPY . /usr/share/nginx/html
EXPOSE 80
# nginx default CMD is fine
EOF
ubuntu@ip-172-31-19-104:~/Azure-Static-Website$ ls
Dockerfile  README.md  index.html
ubuntu@ip-172-31-19-104:~/Azure-Static-Website$
```

i-09882c3c7cbf4144b (DockerDevOps)

Step 5 — Build and Run the Container

```
# Build image
```

```
docker build -t static-site:latest .
```

```
ubuntu@ip-172-31-19-104:~/Azure-Static-Website$ docker build -t static-site:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   85.5kB
Step 1/4 : FROM nginx:alpine
alpine: Pulling from library/nginx
2d35ebdb57d9: Pulling fs layer
0f6a6f33e95d: Pulling fs layer
194fa24e147d: Pulling fs layer
3eaba6cd10a3: Pulling fs layer
df413d6ebdc8: Pulling fs layer
d9a55dab5954: Pulling fs layer
ffa36d5502a: Pulling fs layer
bdabb0d44271: Pulling fs layer
3eaba6cd10a3: Waiting
df413d6ebdc8: Waiting
d9a55dab5954: Waiting
ffa36d5502a: Waiting
bdabb0d44271: Waiting
194fa24e147d: Verifying Checksum
194fa24e147d: Download complete
0f6a6f33e95d: Verifying Checksum
0f6a6f33e95d: Download complete
3eaba6cd10a3: Verifying Checksum
3eaba6cd10a3: Download complete
2d35ebdb57d9: Verifying Checksum
2d35ebdb57d9: Download complete
d9a55dab5954: Verifying Checksum
d9a55dab5954: Download complete
df413d6ebdc8: Verifying Checksum
df413d6ebdc8: Download complete
ffa36d5502a: Verifying Checksum
ffa36d5502a: Download complete
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bdabb0d44271: Verifying Checksum
bdabb0d44271: Download complete
194fa24e147d: Pull complete
3eaba6cd10a3: Pull complete
d9a55dab5954: Pull complete
ffa36d5502a: Pull complete
bdabb0d44271: Pull complete
Digest: sha256:b3c65d5d7ad751196f21b7fd0ebd4da9cb430e32f66adcf92441b72702b14
Status: Downloaded newer image for nginx:alpine
--> d4918ca78576
Step 2/4 : RUN rm -rf /usr/share/nginx/html/*
--> Running in 2ac17e20ef91
--> Removed intermediate container 2ac17e20ef91
--> a7653be098fe
Step 3/4 : COPY . /usr/share/nginx/html
--> 702865ac1f49
Step 4/4 : EXPOSE 80
--> Running in a30406d15656
--> Removed intermediate container a30406d15656
--> 148c7fc726f0
Successfully built 148c7fc726f0
Successfully tagged static-site:latest
ubuntu@ip-172-31-19-104:~/Azure-Static-Website$
```

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Run on port 80

docker run -d --name static-site \

-p 80:80 \

--restart unless-stopped \

static-site:latest

```
ubuntu@ip-172-31-19-104:~/Azure-Static-Website$ docker run -d --name static-site \
-p 80:80 \
--restart unless-stopped \
static-site:latest
d9047484dcea4fe416ba3f521ee47671d8bcb59ebdada5eca80003a69b97d25d
ubuntu@ip-172-31-19-104:~/Azure-Static-Website$
```

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Confirm

docker ps

```
ubuntu@ip-172-31-19-104:~/Azure-Static-Website$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
d9047484dcea   static-site:latest "/docker-entrypoint..." About a minute ago Up About a minute 0.0.0.0:80->80/tcp, [::]:80->80/tcp static-site

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```

Step 6 — Test in Browser

Open: <http://54.145.15.193:80/> (Use the VM's public IP. Ensure NSG/Security Group allows port 80.)

DevOps Micro Internship (#DMI) Cohort 2

by Pravin Mishra

This will be an upgraded one:

- Weekly Live Class (8 hours)** — *lab-first, less lectures.*
 - 1 lunch + 2 coffee/tea breaks
 - Attendance taken **3 times** (start, mid, end)
 - Expect **100% attendance** and **active engagement**
 - Outcome-first:** each sprint ends with a deliverable to post on LinkedIn

Unexcused absence without prior notice → removal from cohort
- Industry-standard projects** — *you ship every week.*
 - 4-6 assignments per week
 - On-time → **manually reviewed & scored**
 - Community + co-mentor support:** you're not alone
 - Maintain **≥ 90% average** (one grace re-submission)

Consistently below threshold → removed from cohort
- Industry-expert mentor support** — *practitioners, not theorists.*

Assignment 44: Multi-Stage Docker Build for a React App (Builder → Nginx Runtime)

1. Clone the React App repository on your VM.

```
ubuntu@ip-172-31-19-104:~$ git clone https://github.com/pravinmishraaws/my-react-app.git
Cloning into 'my-react-app'...
remote: Enumerating objects: 69, done.
remote: Counting objects: 100% (69/69), done.
remote: Compressing objects: 100% (66/66), done.
remote: Total 69 (delta 28), reused 26 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (69/69), 188.95 KiB | 8.59 MiB/s, done.
Resolving deltas: 100% (28/28), done.
ubuntu@ip-172-31-19-104:~$ ls
Azure-Static-Website  my-react-app
ubuntu@ip-172-31-19-104:~$
```

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2. Modify the App.js file to record your name and date.

```
ubuntu@ip-172-31-19-104:~/my-react-app/src$ nano App.js
```

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```
GNU nano 7.2
import React from "react";

function App() {
  return (
    <div style={{ textAlign: "center", paddingTop: "50px" }}>
      <h1>Welcome to My React App</h1>
      <p>This app is running on Nginx!</p>

      <h2>Deployed by: <strong>Adrina Colaco</strong></h2>
      <p>Date: <strong>09-Nov-2025</strong></p>

      <hr style={{ margin: "20px 0" }} />

      <p>
        P.S. This post is part of the FREE DevOps for Beginners Cohort run by
        <strong> Pravin Mishra</strong>.
      </p>
      <p>
        You can start your DevOps journey for free from his{" "}
        <a
          href="https://www.youtube.com/playlist?list=PLVodqXbCs7bX88JeUZmK4fKTq2h3"
          target="_blank"
          rel="noopener noreferrer"
        >
          YouTube Playlist
        </a>.
      </p>
    </div>
  );
}
```

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3. Create Dockerfile

```
ubuntu@ip-172-31-19-104:~/my-react-app$ vi Dockerfile
ubuntu@ip-172-31-19-104:~/my-react-app$
```

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```
# Use the official Node.js image as the base image
FROM node:18-alpine AS build

# Set the working directory
WORKDIR /app

# Copy package.json and install dependencies
COPY package.json package-lock.json ./
RUN npm install

# Copy the entire project and build the app
COPY . .
RUN npm run build

# Use Nginx to serve the React app
FROM nginx:alpine
COPY --from=build /app/build /usr/share/nginx/html

# Expose port 80 to allow external access
EXPOSE 80

# Start Nginx
CMD ["nginx", "-g", "daemon off;"]

~
~
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~
~
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-- INSERT --
```

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4. Build the image
docker build -t my-react-app .

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```

5. Run the image

```
docker run -d -p 80:80 --name react-container my-react-app
```

```
ubuntu@ip-172-31-19-104:~/my-react-app$ docker run -d -p 80:80 --name react-container my-react-app
b267fee7f9a4b4f5e6f55c304e7d307d335d9f2f8dc89f4c5dde89b12ca08b2
ubuntu@ip-172-31-19-104:~/my-react-app$
```

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6. Check if it exists

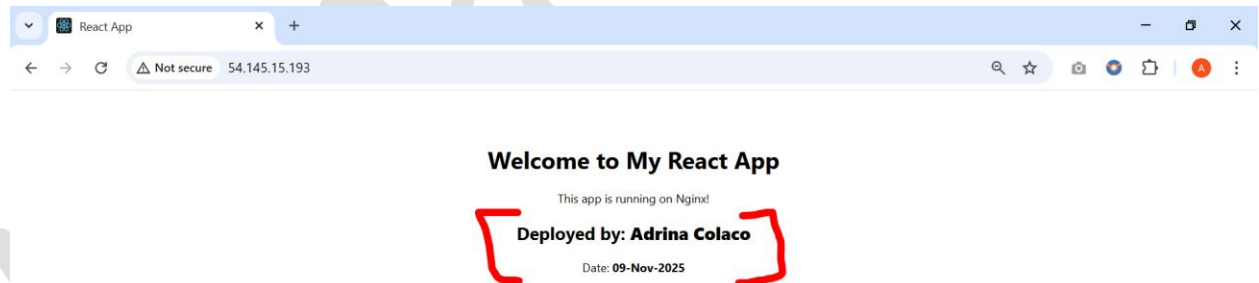
```
docker images
```

```
ubuntu@ip-172-31-19-104:~/my-react-app$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
my-react-app         latest             fc1e2a25b2e9       5 minutes ago      53.9MB
<none>               <none>            72fcd4971a81       5 minutes ago      414MB
static-site          latest             148c7fc726f0       2 days ago         52.9MB
nginx                alpine             d4918ca78576       12 days ago        52.8MB
node                 18-alpine          ee77c6cd7c18       7 months ago       127MB
ubuntu@ip-172-31-19-104:~/my-react-app$
```

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7. Access the application from the container via public IP of the VM.



8. Display logs

```
docker logs -f react-container
```

Check how logs are getting generated at real time even when you refresh your browser page.

```
aws
[Alt+S]
United States (N. Virginia)
Account ID: 3111-4154-2115
Adriana Colaco

ubuntu@ip-172-31-19-104:~$ docker logs -f react-container
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/11/10 14:33:27 [notice] 1#1: using the "epoll" event method
2025/11/10 14:33:27 [notice] 1#1: nginx/1.29.3
2025/11/10 14:33:27 [notice] 1#1: built by gcc 14.2.0 (Alpine 14.2.0)
2025/11/10 14:33:27 [notice] 1#1: OS: Linux 6.14.0-1015-aws
2025/11/10 14:33:27 [notice] 1#1: getrlimit(RLIMIT_Nofile): 1048576:1048576
2025/11/10 14:33:27 [notice] 1#1: start worker processes
2025/11/10 14:33:27 [notice] 1#1: start worker process 30
2025/11/10 14:33:27 [notice] 1#1: start worker process 30
122.169.87.151 - - [10/Nov/2025:14:34:58 +0000] "GET / HTTP/1.1" 200 644 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [10/Nov/2025:14:34:58 +0000] "GET /static/css/main.e6c13ad2.css HTTP/1.1" 200 337 "http://54.145.15.193/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [10/Nov/2025:14:34:59 +0000] "GET /static/js/main.ae2cf689.js HTTP/1.1" 200 186362 "http://54.145.15.193/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [10/Nov/2025:14:37:00 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [10/Nov/2025:14:37:00 +0000] "GET /static/js/main.ae2cf689.js HTTP/1.1" 304 0 "http://54.145.15.193/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [10/Nov/2025:14:37:00 +0000] "GET /static/css/main.e6c13ad2.css HTTP/1.1" 304 0 "http://54.145.15.193/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [10/Nov/2025:14:37:01 +0000] "GET /favicon.ico HTTP/1.1" 200 3870 "http://54.145.15.193/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [10/Nov/2025:14:37:06 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"

i-09882c3c7cbf4144b (DockerDevOps)
PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

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```

Assignment 45: Docker Networking — Connect Frontend ↔ Backend via Custom Bridge (DNS by Name)

1. List Available Networks

docker network ls

```
ubuntu@ip-172-31-19-104:~$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
759f87e91b59        bridge              bridge              local
d805627e0806        host                host                local
2903bd83996b        none                null                local
ubuntu@ip-172-31-19-104:~$
```

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2. PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

To run a standalone application:

3. Check if the Nginx Image is Available Locally

Output should have nginx

```
ubuntu@ip-172-31-19-104:~$ docker images
REPOSITORY    TAG               IMAGE ID           CREATED            SIZE
nginx          alpine            d4918ca78576      12 days ago       52.8MB
node          18-alpine        ee77c6cd7c18      7 months ago      127MB
ubuntu@ip-172-31-19-104:~$
```

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If not present then run
docker search nginx
docker pull nginx

4. Check If Any Containers Are Running

docker ps

```
ubuntu@ip-172-31-19-104:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

```
ubuntu@ip-172-31-19-104:~$
```

i-09882c3c7cbf4144b (DockerDevOps)

PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

5. Run the Nginx Container

docker run -d -p 80:80 --name myweb nginx

```
ubuntu@ip-172-31-19-104:~$ docker run -d -p 80:80 --name myweb nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
d7ecded7702a: Pull complete
266626526d42: Pull complete
320b0949be89: Pull complete
d921c57c6a81: Pull complete
9def903993e4: Pull complete
52bc359bcbd7: Pull complete
e2f8e296d9df: Pull complete
Digest: sha256:1beed3ca46acebe9d3fb62e9067f03d05d5bfa97a00f30938a0a3580563272ad
Status: Downloaded newer image for nginx:latest
217803db2ff52ccaca4408b282605a9cf718f5dbc37abfa970006766f0182841
ubuntu@ip-172-31-19-104:~$
```

i-09882c3c7cbf4144b (DockerDevOps)

PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

6. Verify the Container is Running

docker ps

```
ubuntu@ip-172-31-19-104:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
217803db2ff5	nginx	"/docker-entrypoint..."	2 minutes ago	Up 2 minutes	0.0.0.0:80->80/tcp, [::]:80->80/tcp	myweb

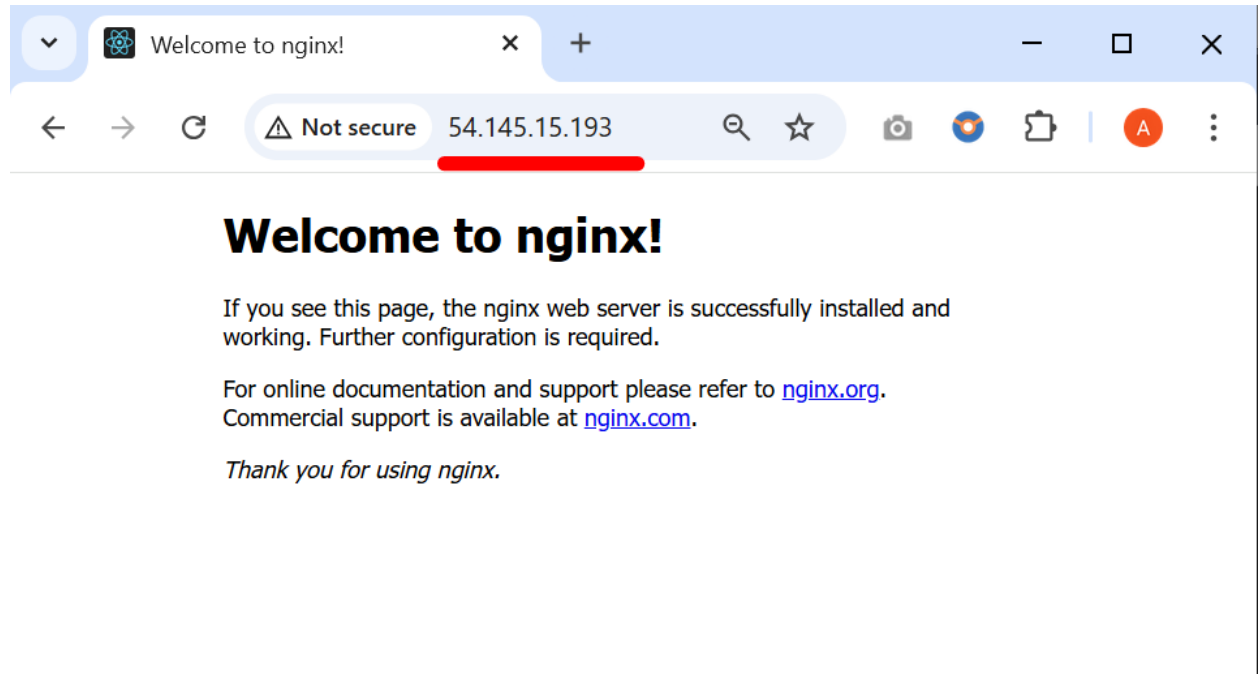
```
ubuntu@ip-172-31-19-104:~$
```

i-09882c3c7cbf4144b (DockerDevOps)

PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

7. Access the application

<http://publicIP>



Two Containers Communicating (Microservices)

8. Cleanup Previous Containers and Images

docker ps

```
ubuntu@ip-172-31-19-104:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                    NAMES
217803db2ff5   nginx    "/docker-entrypoint..." About an hour ago Up About an hour   0.0.0.0:80->80/tcp, [::]:80->80/tcp   myweb
ubuntu@ip-172-31-19-104:~$
```

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PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

docker stop myweb

docker rm myweb

```
ubuntu@ip-172-31-19-104:~$ docker stop myweb
myweb
ubuntu@ip-172-31-19-104:~$ docker rm myweb
myweb
ubuntu@ip-172-31-19-104:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                    NAMES
ubuntu@ip-172-31-19-104:~$
```

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PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

docker images

docker rmi nginx

```
ubuntu@ip-172-31-19-104:~$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    d261fd19cb63   6 days ago    152MB
nginx         alpine    d4918ca78576   12 days ago   52.8MB
node          18-alpine ee77c6cd7c18   7 months ago  127MB
ubuntu@ip-172-31-19-104:~$ docker rmi nginx
Untagged: nginx:latest
Untagged: nginx@sha256:1beed3ca46acebe9d3fb62e9067f03d05d5bfa97a00f30938a0a3580563272ad
Deleted: sha256:d261fd19cb63238535ab80d4e1be1d9e7f6c8b5a28a820188968dd3e6f06072d
Deleted: sha256:534fa102b912db29b236581b36d96b3961fab8a8e2f87170de3a97444075880e
Deleted: sha256:f356d7e216ceca7046179c3ac20a975853d782fe5f914fd76275ba1949ae7e5b
Deleted: sha256:f8d71bebdd1664ec5c412cc511bde3a5a93df38254ac3e59431da4e9830b3ffa
Deleted: sha256:d65f0afc2c55c237e5a061e22de0cf6f8b24c171b89902f0e90675d50ecce0aa
Deleted: sha256:8edab3f21d7cfc317f5e932991903851eddf379ab6c50cf806675a63cdd91d21
Deleted: sha256:813c6b1f7312b3773f41f7bee85244570fed62c2aa5e7ba45d83d7fc16106e2c
Deleted: sha256:36d06fe0cbc654e5f67d58c960ed33e53127e4a3288d8ce6f6a60a9c311794d4
ubuntu@ip-172-31-19-104:~$
```

i-09882c3c7cbf4144b (DockerDevOps)

PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

9. Create a Custom Network

docker network create mynetwork

docker network ls

```
ubuntu@ip-172-31-19-104:~$ docker network create mynetwork
2348389428de4e6f6b0d3e6a81829f8f7a0b840014e5eaf82ae83f1dbf1285e5
ubuntu@ip-172-31-19-104:~$ docker network ls
NETWORK ID        NAME          DRIVER  SCOPE
759f87e91b59     bridge       bridge  local
d805627e0806     host         host    local
2348389428de     mynetwork    bridge  local
2903bd83996b     none         null    local
ubuntu@ip-172-31-19-104:~$
```

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10. Run the frontend application

Create a Dockerfile for frontend

```
ubuntu@ip-172-31-19-104:~/Networking$ nano dockerfile
ubuntu@ip-172-31-19-104:~/Networking$
```

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```

GNU nano 7.2                                                                 dockerfile *
# Use the official Node.js 18 image
FROM node:18

# Set the working directory inside the container
WORKDIR /app

# Initialize a new Node.js app and install Express
RUN npm init -y && npm install express

# Create index.js with basic Express code running on port 80
RUN echo "const express = require('express'); \
const app = express(); \
app.get('/', (req, res) => res.send('Hello from Frontend')); \
const PORT = 80; \
app.listen(PORT, () => console.log('Frontend running on port ' + PORT));" > index.js

# Expose port 80 to the host
EXPOSE 80

# Run the application
CMD ["node", "index.js"]

```

11. Build and run the Frontend Container

`docker build -t frontend-app .`

```

aws [Alt+S] United States (N. Virginia) Account ID: 3111-4154-2113 Adrina Colaco
ubuntu@ip-172-31-19-104:~/Networking$ docker build -t frontend-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.56kB
Step 1/6 : FROM node:18
--> b50082bc3670
Step 2/6 : WORKDIR /app
--> Using cache
--> 67ff1b654ba5
Step 3/6 : RUN npm init -y && npm install express
--> Running in d3fb8e26ac2a
Wrote to /app/package.json:

{
  "name": "app",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

added 68 packages, and audited 69 packages in 4s
16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
--> Removed intermediate container d3fb8e26ac2a
Step 4/6 : RUN echo "const express = require('express'); const app = express(); app.get('/', (req, res) => res.send('Hello from Frontend')); const PORT = 80; app.listen(PORT, () => console.log('Frontend running on port ' + PORT));" > index.js
--> Running in bb7afe9b52d
--> Removed intermediate container bb7afe9b52d
--> 304573cfd54
Step 5/6 : EXPOSE 80
--> Running in 57ed5ef9f273
--> Removed intermediate container 57ed5ef9f273
--> 5ad409591ba2
Step 6/6 : CMD ["node", "index.js"]
--> Running in 026f2fceb849
--> Removed intermediate container 026f2fceb849
--> 1408eca009b8
Successfully built 1408eca009b8
Successfully tagged frontend-app:latest
ubuntu@ip-172-31-19-104:~/Networking$

i-09882c3c7cbf4144b (DockerDevOps)
PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

```

`docker run -d --name frontend --network mynetwork -p 80:80 frontend-app`

```

ubuntu@ip-172-31-19-104:~/Networking$ docker run -d --name frontend --network mynetwork -p 80:80 frontend-app
c9163601b1b2549ba9008e62ffb607e672d50b33770a8919345b8ed0472eb650
ubuntu@ip-172-31-19-104:~/Networking$

```

i-09882c3c7cbf4144b (DockerDevOps)

PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

12. Verify container is running

```
ubuntu@ip-172-31-19-104:~/Networking$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
c9163601b1b2	frontend-app	"docker-entrypoint.s..."	About a minute ago	Up About a minute	0.0.0.0:80->80/tcp, [::]:80->80/tcp	frontend

```
ubuntu@ip-172-31-19-104:~/Networking$
```

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13. Testing backend from the host

curl http://<publicIP>

```
ubuntu@ip-172-31-19-104:~/Networking$ curl http://54.145.15.193
```

Hello from Frontendubuntu@ip-172-31-19-104:~/Networking\$

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PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

14. Run the Backend

docker run -d --name backend --network mynetwork nginx

```
Hello from Frontendubuntu@ip-172-31-19-104:~/Networking$ docker run -d --name backend --network mynetwork nginx
```

Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
d7ecded7702a: Pull complete
266626526d42: Pull complete
320b0949be89: Pull complete
d921c57c6a81: Pull complete
9def903993e4: Pull complete
52bc359bcbd7: Pull complete
e2f8e296d9df: Pull complete
Digest: sha256:1beed3ca46acebe9d3fb62e9067f03d05d5bfa97a00f30938a0a3580563272ad
Status: Downloaded newer image for nginx:latest
fb1f17be3e4afabd3ac5af8d950e29588083c6915760fbdb8e3fe0025ba4b4b
ubuntu@ip-172-31-19-104:~/Networking\$

i-09882c3c7cbf4144b (DockerDevOps)
PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

docker ps

```
ubuntu@ip-172-31-19-104:~/Networking$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
fb1f17be3e4a	nginx	"/docker-entrypoint.s..."	About a minute ago	Up About a minute	80/tcp	backend
c9163601b1b2	frontend-app	"docker-entrypoint.s..."	9 minutes ago	Up 9 minutes	0.0.0.0:80->80/tcp, [::]:80->80/tcp	frontend

```
ubuntu@ip-172-31-19-104:~/Networking$
```

i-09882c3c7cbf4144b (DockerDevOps)
PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

15. Verify network connectivity

docker network inspect mynetwork

```
awscli Search [Alt+S] United States (N. Virginia) Account ID: 3111-4154-2113 Adrina Colaco

ubuntu@ip-172-31-19-104:~/Networking$ docker network inspect mynetwork
[
  {
    "Name": "mynetwork",
    "Id": "2348389428de4e6f6bd3e6a81829f8f7a0b840014e5eaf82ae83f1dbf1285e5",
    "Created": "2025-11-10T17:41:28.497660997Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": [
      {
        "c9163601b1b2549ba9008e62ffb607e672d50b33770a8919345b8ed0472eb650": {
          "Name": "frontend",
          "EndpointID": "a830cc3417aa8814916dde8dba542cfd29ba657a680eeb27e700c9e52b83dbe3",
          "MacAddress": "6e:54:93:9c:27:e2",
          "IPv4Address": "172.18.0.2/16",
          "IPv6Address": ""
        },
        {
          "fb1f17bc3efafbd3ac5af8d950e29588083c6915760fbedb8e3fe0025ba4b4b": {
            "Name": "backend",
            "EndpointID": "40516fc24e4b8785dd0ca7d1a49c799295f80de56679ea974cc47f83189d60c5",
            "MacAddress": "62:1b:c7:72:20:21",
            "IPv4Address": "172.18.0.3/16",
            "IPv6Address": ""
          }
        }
      ],
      "Options": {},
      "Labels": {}
    }
  ]
}

ubuntu@ip-172-31-19-104:~/Networking$

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PublicIPs: 54.145.15.193 PrivateIPs: 172.31.19.104

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```

16. Test communication Between Frontend and Backend

a. Test Manually

docker exec -it frontend apt update && docker exec -it frontend apt install curl -y

docker exec -it frontend curl backend

```
aws
Search [Alt+S]
United States (N. Virginia) Account ID: 3111-4154-2113 Adrina Colaco

ubuntu@ip-172-31-19-104:~/Networking$ docker exec -it frontend apt update && docker exec -it frontend apt install curl -y
docker exec -it frontend curl backend
Get:1 http://deb.debian.org/debian bookworm InRelease [151 kB]
Get:2 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:3 http://deb.debian.org/debian-security bookworm-security InRelease [48.0 kB]
Get:4 http://deb.debian.org/debian bookworm/main amd64 Packages [8791 kB]
Get:5 http://deb.debian.org/debian bookworm-updates/main amd64 Packages [6924 B]
Get:6 http://deb.debian.org/debian-security bookworm-security/main amd64 Packages [284 kB]
Fetched 9337 kB in 2s (5873 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
102 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl3-gnutls libcurl4 libcurl4-openssl-dev
Suggested packages:
  libcurl4-doc libidn-dev libldap2-dev librtmp-dev libssh2-1-dev
The following packages will be upgraded:
  curl libcurl3-gnutls libcurl4 libcurl4-openssl-dev
4 upgraded, 0 newly installed, 0 to remove and 98 not upgraded.
Need to get 1586 kB of archives.
After this operation, 1024 B of additional disk space will be used.
Get:1 http://deb.debian.org/debian bookworm/main amd64 libcurl4-openssl-dev amd64 7.88.1-10+deb12u14 [492 kB]
Get:2 http://deb.debian.org/debian bookworm/main amd64 curl amd64 7.88.1-10+deb12u14 [316 kB]
Get:3 http://deb.debian.org/debian bookworm/main amd64 libcurl4 amd64 7.88.1-10+deb12u14 [392 kB]
Get:4 http://deb.debian.org/debian bookworm/main amd64 libcurl3-gnutls amd64 7.88.1-10+deb12u14 [386 kB]
Fetched 1586 kB in 0s (35.7 MB/s)
debconf: delaying package configuration, since apt-utils is not installed
(Reading database ... 23250 files and directories currently installed.)
Preparing to unpack .../libcurl4-openssl-dev_7.88.1-10+deb12u14_amd64.deb ...
Unpacking libcurl4-openssl-dev:amd64 (7.88.1-10+deb12u14) over (7.88.1-10+deb12u12) ...
Preparing to unpack .../curl_7.88.1-10+deb12u14_amd64.deb ...
Unpacking curl (7.88.1-10+deb12u14) over (7.88.1-10+deb12u12) ...
Preparing to unpack .../libcurl4_7.88.1-10+deb12u14_amd64.deb ...
Unpacking libcurl4:amd64 (7.88.1-10+deb12u14) over (7.88.1-10+deb12u12) ...
Preparing to unpack .../libcurl3-gnutls_7.88.1-10+deb12u14_amd64.deb ...
Unpacking libcurl3-gnutls:amd64 (7.88.1-10+deb12u14) over (7.88.1-10+deb12u12) ...
Setting up libcurl4:amd64 (7.88.1-10+deb12u14) ...
Setting up curl (7.88.1-10+deb12u14) ...
Setting up libcurl3-gnutls:amd64 (7.88.1-10+deb12u14) ...
Processing triggers for libc-bin (2.36-9+deb12u10) ...
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p>Thank you for using nginx.</p>
</body>
</html>
ubuntu@ip-172-31-19-104:~/Networking$
```

- b. Enter Frontend Container
Access the Frontend Container
docker exec -it frontend /bin/sh
apt update && apt install curl -y
curl backend
exit

```

ubuntu@ip-172-31-19-104:~/Networking$ docker exec -it frontend /bin/sh
# apt update && apt install curl -y
Hit:1 http://deb.debian.org/debian bookworm InRelease
Hit:2 http://deb.debian.org/debian bookworm-updates InRelease
Hit:3 http://deb.debian.org/debian-security bookworm-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
98 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (7.88.1-10+deb12u14).
0 upgraded, 0 newly installed, 0 to remove and 98 not upgraded.
# curl backend
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
# exit

```

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Assignment 46: Docker Data Persistence — Bind Mounts vs Volumes (Logs + Shared Storage)

Scenario A — Bind Mounts for Persistent Logs

1. Prepare a clean workspace on a VM or local machine with Docker installed.

```
ubuntu@ip-172-31-19-104:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
ubuntu@ip-172-31-19-104:~$
```

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2. Create a **host directory** dedicated to web logs (e.g., a folder for access/error logs).

```
mkdir -p ~/docker_logs/web
```

```
ubuntu@ip-172-31-19-104:~$ mkdir -p ~/docker_logs/nginx
ubuntu@ip-172-31-19-104:~$ cd docker_logs/
ubuntu@ip-172-31-19-104:~/docker_logs$ ls
nginx
ubuntu@ip-172-31-19-104:~/docker_logs$
```

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3. Run a web server container that:

- Exposes HTTP to your host for browser testing.
- **Mounts** the host log directory to the container's log path (so logs are written on the host).

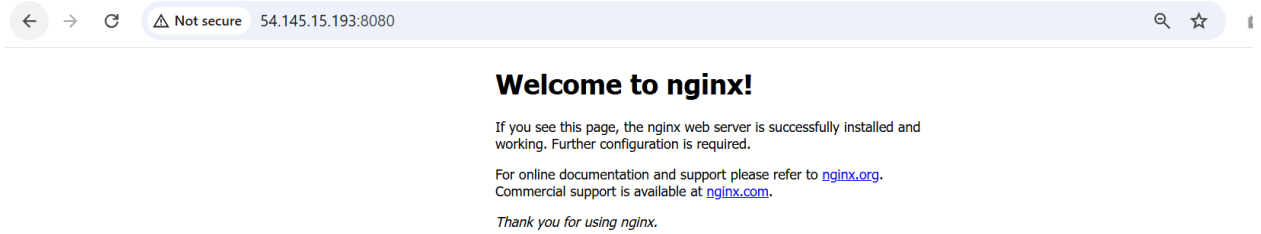
```
docker run -d --name web-bind-nginx -p 8080:80 -v ~/docker_logs/nginx:/var/log/nginx nginx:latest
```

```
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$ docker run -d --name web-bind-nginx -p 8080:80 -v ~/docker_logs/nginx:/var/log/nginx nginx:latest
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
d7ecded7702a: Pull complete
266626526d42: Pull complete
320b0949be89: Pull complete
4921c597c6a01: Pull complete
9def903993a4: Pull complete
52bc359bcb47: Pull complete
e2f8e296d9df: Pull complete
Digest: sha256:1beed3ca46acebe9d3fb62e9067f03d05d5bfa97a00f30938a0a3580563272ad
Status: Downloaded newer image for nginx:latest
d27b0443763afc553ff0e660a61b4b512d0b0758e16c2975d24577f37ef66a0b
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$
```

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- Open the site in a browser a few times to generate traffic.
Add port 8080 to the security group of the VM.



- Verify that the **access** and **error** log files are created on the **host** and include new entries.

```
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$ ls
access.log  error.log
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$
```

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```
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$ cat access.log
122.169.87.151 - - [11/Nov/2025:08:10:52 +0000] "GET / HTTP/1.1" 200 415 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [11/Nov/2025:08:10:53 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://54.145.15.193:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [11/Nov/2025:08:11:30 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [11/Nov/2025:08:11:32 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$

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ubuntu@ip-172-31-19-104:~/docker_logs/nginx$ cat error.log
2025/11/11 07:26:37 [notice] 181: using the "epoll" event method
2025/11/11 07:26:37 [notice] 181: nginx/1.29.3
2025/11/11 07:26:37 [notice] 181: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/11/11 07:26:37 [notice] 181: OS: Linux 6.14.0-1015-aws
2025/11/11 07:26:37 [notice] 181: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/11/11 07:26:37 [notice] 181: start worker processes
2025/11/11 07:26:37 [notice] 181: start worker process 30
2025/11/11 07:43:43 [notice] 181: signal 3 (SIGQUIT) received, shutting down
2025/11/11 07:43:43 [notice] 30#30: gracefully shutting down
2025/11/11 07:43:43 [notice] 30#30: exiting
2025/11/11 07:43:43 [notice] 30#30: exit
2025/11/11 07:43:43 [notice] 181: signal 17 (SIGCHLD) received from 30
2025/11/11 07:43:43 [notice] 181: worker process 30 exited with code 0
2025/11/11 07:43:43 [notice] 181: exit
2025/11/11 07:46:31 [notice] 181: using the "epoll" event method
2025/11/11 07:46:31 [notice] 181: nginx/1.29.3
2025/11/11 07:46:31 [notice] 181: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/11/11 07:46:31 [notice] 181: OS: Linux 6.14.0-1015-aws
2025/11/11 07:46:31 [notice] 181: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/11/11 07:46:31 [notice] 181: start worker processes
2025/11/11 07:46:31 [notice] 181: start worker process 29
2025/11/11 08:10:53 [error] 29#29: *3 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 122.169.87.151, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "54.145.15.193:8080", referer: "http://54.145.15.193:8080/"
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$
```

- Remove the container. Confirm the **log** files **remain** on the host unchanged.

```
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$ docker rm -f web-bind-nginx
web-bind-nginx
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$ ls
access.log  error.log
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$
```

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7. Start a **new** web container pointing to the **same** host log directory.

```
docker run -d --name web-bind-nginx1 -p 8080:80 -v ~/docker_logs/nginx:/var/log/nginx
nginx:latest
```

```
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$ docker run -d --name web-bind-nginx1 -p 8080:80 -v ~/docker_logs/nginx:/var/log/nginx nginx:latest
e80a6c5ea5996abde4e5bcd6a3f1e63e438b16772e1274dbff74e64d8837b66c
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$
```

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8. Hit the site again and confirm the **same** host log files now include **new entries** (continuous history).

```
cat ~/docker_logs/nginx/access.log
```

```
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$ cat access.log
122.169.87.151 - - [11/Nov/2025:08:10:53 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [11/Nov/2025:08:10:53 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://54.145.15.193:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [11/Nov/2025:08:11:30 +0000] "GET / HTTP/1.1" 204 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [11/Nov/2025:08:11:32 +0000] "GET / HTTP/1.1" 204 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [11/Nov/2025:08:15:29 +0000] "GET / HTTP/1.1" 204 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
122.169.87.151 - - [11/Nov/2025:08:15:59 +0000] "GET / HTTP/1.1" 204 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36" "-"
ubuntu@ip-172-31-19-104:~/docker_logs/nginx$
```

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Observations & Notes

- Using NGINX, logs by default go to /var/log/nginx/access.log and /var/log/nginx/error.log inside the container.
- Risk of bind mounts remains: your host directory structure is tightly coupled to container configuration; moving to a different host may require re-setup; permissions or SELinux issues may surface.
- Note: The official NGINX Docker image sometimes redirects logs to stdout/stderr via symlinks: /var/log/nginx/access.log → /dev/stdout, /var/log/nginx/error.log → /dev/stderr.
 - If you use bind mount, you override that behavior and ensure log files exist on host FS.

Scenario B — Named Volume for Shared Data (Writer/Reader)

1. Cleanup Previous Containers and Images

```
docker ps
docker stop myweb
docker rm myweb
docker images
docker rmi nginx
```

2. Set Up the Folder Structure

```
mkdir -p two-tier-app/{frontend,backend}
```

```
ubuntu@ip-172-31-19-104:~$ mkdir -p two-tier-app/{frontend,backend}
ubuntu@ip-172-31-19-104:~$ cd two-tier-app/
ubuntu@ip-172-31-19-104:~/two-tier-app$ ls
backend frontend
ubuntu@ip-172-31-19-104:~/two-tier-app$
```

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```
ubuntu@ip-172-31-19-104:~/two-tier-app$ cd backend/
ubuntu@ip-172-31-19-104:~/two-tier-app/backend$ touch Dockerfile
ubuntu@ip-172-31-19-104:~/two-tier-app/backend$ cd ..
ubuntu@ip-172-31-19-104:~/two-tier-app$ cd frontend/
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$ touch Dockerfile
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$
```

3. Create a Custom Docker Network

```
docker network create mynetwork
```

```
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$ docker network create mynetwork
441fa74550a5576821f5fbcb6fc9eed9a758ec25437ba864b6bda25685eb949e
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$
```

4. Create a Shared Volume for Data Persistence

docker volume create shared-data

docker volume ls

```
ubuntu@ip-172-31-19-104:~/two-tier-app$ docker volume create shared-data
docker volume ls
shared-data
DRIVER      VOLUME NAME
local       shared-data
ubuntu@ip-172-31-19-104:~/two-tier-app$
```

5. Build and Run the Backend Service

- Create the Backend Dockerfile
- **backend (writer)**: on a specific HTTP path (e.g., /write), it writes/updates **/data/message.txt** in the **mounted named volume**.

```
GNU nano 7.2 Dockerfile
# Use the official Node.js image
FROM node:18

# Set the working directory inside the container
WORKDIR /app

# Install required dependencies
RUN npm init -y && npm install express fs

# Create index.js with API logic
RUN echo "const express = require('express'); \
const fs = require('fs'); \
const app = express(); \
const DATA_FILE = '/data/message.txt'; \
app.get('/write', (req, res) => { fs.writeFileSync(DATA_FILE, 'Hello from Backend!'); res.send('Data written!'); }); \
app.listen(80, () => console.log('Backend running on port 80'));" > index.js

# Expose port 80 to the host
EXPOSE 80

# Run the application
CMD ["node", "index.js"]
```

- Build and Run the Backend Container
docker build -t backend-app .

```
aws Search [Alt+S] United States (N. Virginia) Account ID: 3111-4154-2113 Adria Colaco

ubuntu@ip-172-31-19-104:~/two-tier-app/backend$ docker build -t backend-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.56kB
Step 1/6 : FROM node:18
18: Pulling from library/node
3e6b9d1a9511: Pulling fs layer
37927e4901b1: Pulling fs layer
79b2f47ad444: Pulling fs layer
c23f099911d6: Pulling fs layer
cda7f44f2bd4: Pulling fs layer
c6b30c3f1696: Pulling fs layer
3697be50c98b: Pulling fs layer
461077a72fb7: Pulling fs layer
e23f099911d6: Waiting
cda7f44f2bd4: Waiting
c6b30c3f1696: Waiting
461077a72fb7: Waiting
3697be50c98b: Waiting
37927e4901b1: Verifying Checksum
37927e4901b1: Download complete
3e6b9d1a9511: Verifying Checksum
3e6b9d1a9511: Download complete
cda7f44f2bd4: Verifying Checksum
cda7f44f2bd4: Download complete
79b2f47ad444: Verifying Checksum
79b2f47ad444: Download complete
3697be50c98b: Verifying Checksum
3697be50c98b: Download complete
461077a72fb7: Verifying Checksum
461077a72fb7: Download complete
c6b30c3f1696: Verifying Checksum
c6b30c3f1696: Download complete
e23f099911d6: Verifying Checksum
e23f099911d6: Download complete
3e6b9d1a9511: Pull complete
37927e4901b1: Pull complete
79b2f47ad444: Pull complete
e23f099911d6: Pull complete
cda7f44f2bd4: Pull complete
c6b30c3f1696: Pull complete
3697be50c98b: Pull complete
461077a72fb7: Pull complete
Digest: sha256:cda7f44f2bd479b2f47ad444c23f099911d6cda7f44f2bd479b2f47ad444c23f099911d6
Status: Downloaded newer image for node:18
--> b5082b2c3670
Step 2/6 : WORKDIR /app
--> Running in ace2e972e58e
--> Removed intermediate container ace2e972e58e
--> 489715c93142
Step 3/6 : RUN npm init -y && npm install express fs
--> Running in d8037e50f6ce
wrote to /app/package.json:

{
  "name": "app",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

added 69 packages, and audited 70 packages in 4s
16 packages are looking for funding
  run 'npm fund' for details

found 0 vulnerabilities
--> Removed intermediate container d8037e50f6ce
--> 4715f4e484de
Step 4/6 : RUN echo "const express = require('express'); const fs = require('fs'); const app = express(); const DATA_FILE = '/data/message.txt'; app.get('/', (req, res) => { fs.writeFileSync(DATA_FILE, 'Hello from Backend!'); res.send('Data written!'); }); app.listen(80, () => console.log('Backend running on port 80'));" > index.js
--> Running in 3bc222fc843b
--> Removed intermediate container 3bc222fc843b
--> 056ab6ef04ee
Step 5/6 : EXPOSE 80
--> Running in e6ae92dc81b2
--> Removed intermediate container e6ae92dc81b2
--> 6ea3980d03ec
Step 6/6 : CMD ["node", "index.js"]
--> Running in 34dfe09f48c3
--> Removed intermediate container 34dfe09f48c3
--> cf64b7b5f8e0
Successfully built cf64b7b5f8e0
Successfully tagged backend-app:latest
ubuntu@ip-172-31-19-104:~/two-tier-app/backend$
```

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docker run -d --name backend --network mynetwork -v shared-data:/data
backend-app

```
ubuntu@ip-172-31-19-104:~/two-tier-app/backend$ docker run -d --name backend --network mynetwork -v shared-data:/data backend-app
e1b7e9ef5105924594fc5b42725c6cd4c86b4858279fe4f3a7ae7b4f4e44fa0b
ubuntu@ip-172-31-19-104:~/two-tier-app/backend$
```

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docker ps

```
ubuntu@ip-172-31-19-104:~/two-tier-app/backend$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
e1b7e9ef5105   backend-app    "docker-entrypoint.s..." 21 seconds ago Up 20 seconds 80/tcp         backend
```

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6. Build and Run the Frontend Service

- Create the Frontend Dockerfile
- **frontend (reader)**: on its root path, it reads **/data/message.txt** from the **same named volume** and returns the content; if missing, returns a clear “no data” message.

```
GNU nano 7.2 Dockerfile
# Use the official Node.js image
FROM node:18

# Set the working directory inside the container
WORKDIR /app

# Install required dependencies
RUN npm init -y && npm install express fs

# Create index.js to serve data
RUN echo "const express = require('express'); \
const fs = require('fs'); \
const app = express(); \
const DATA_FILE = '/data/message.txt'; \
app.get('/', (req, res) => { res.send(fs.existsSync(DATA_FILE) ? fs.readFileSync(DATA_FILE, 'utf8') : 'No data found.')}); \
app.listen(80, () => console.log('Frontend running on port 80'));" > index.js

# Expose port 80 to the host
EXPOSE 80

# Run the application
CMD ["node", "index.js"]
```

- Build and Run the Frontend Container
- docker build -t frontend-app

```
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$ docker build -t frontend-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with Buildkit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.56kB
Step 1/6 : FROM node:18
--> b50082bc3670
Step 2/6 : WORKDIR /app
--> 440715c93142
Step 3/6 : RUN npm init -y && npm install express fs
--> a71f5f4de9da
Step 4/6 : RUN echo "const express = require('express'); const fs = require('fs'); const app = express(); const DATA_FILE = '/data/message.txt'; app.get('/', (req, res) => { res.send(fs.existsSync(DATA_FILE) ? fs.readFileSync(DATA_FILE, 'utf8') : 'No data found.')}); app.listen(80, () => console.log('Frontend running on port 80'));" > index.js
--> 9bde9959658b
--> Removed intermediate container 9bde9959658b
Step 5/6 : EXPOSE 80
--> Running in 86993b8ed2ae
--> Removed intermediate container 86993b8ed2ae
Step 6/6 : CMD ["node", "index.js"]
--> Running in 5bec9e855462
--> Removed intermediate container 5bec9e855462
--> 7db4ca30f26e
Successfully built 7db4ca30f26e
Successfully tagged frontend-app:latest
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$
```

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```
docker run -d --name frontend --network mynetwork -v shared-data:/data -p 80:80 frontend-app
```

```
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$ docker run -d --name frontend --network mynetwork -v shared-data:/data -p 80:80 frontend-app
88d7d93e5452d1ee3199c1ca762d540b0882c59658a7100d7cf0e0323ce3293
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$
```

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docker ps

```
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
88d7d93e5452   frontend-app   "docker-entrypoint.s..." 21 seconds ago Up 21 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp frontend
e1b7e9ef5105   backend-app    "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  80/tcp                             backend
ubuntu@ip-172-31-19-104:~/two-tier-app/frontend$
```

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7. Test Data Sharing Between Backend and Frontend

Test the flow:

- Write Data from the Backend

docker exec backend curl <http://localhost/write>

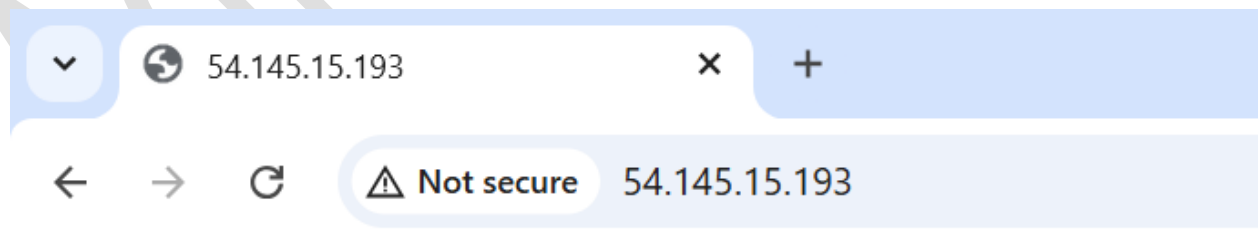
```
ubuntu@ip-172-31-19-104:~/two-tier-app$ docker exec backend curl http://localhost/write
% Total % Received % Xferd Average Speed Time Time Time Current
         Dload Upload   Total   Spent    Left   Speed
100 Data written! 13 100 13 0 0 729 0 --:--:-- --:--:-- --:--:-- 764
ubuntu@ip-172-31-19-104:~/two-tier-app$
```

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- Read Data from the Frontend

<http://<PublicIP>>



Hello from Backend!

- Manually Modify Data Multiple Times

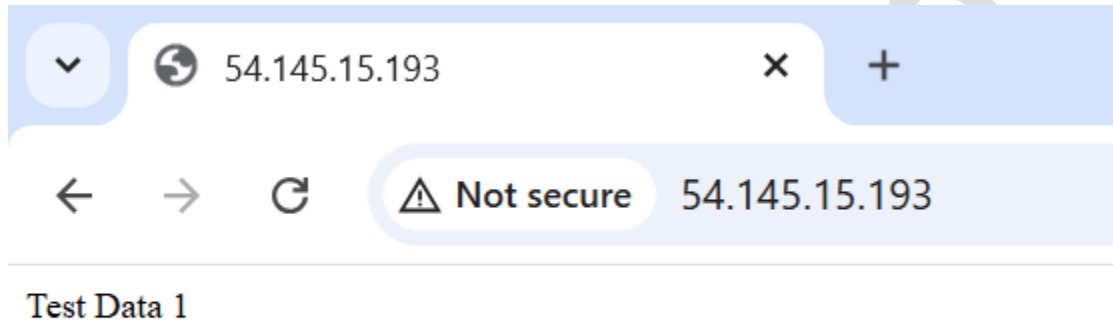
`docker exec backend sh -c "echo 'Test Data 1' > /data/message.txt"`

```
ubuntu@ip-172-31-19-104:~/two-tier-app$ docker exec backend sh -c "echo 'Test Data 1' > /data/message.txt"
ubuntu@ip-172-31-19-104:~/two-tier-app$
```

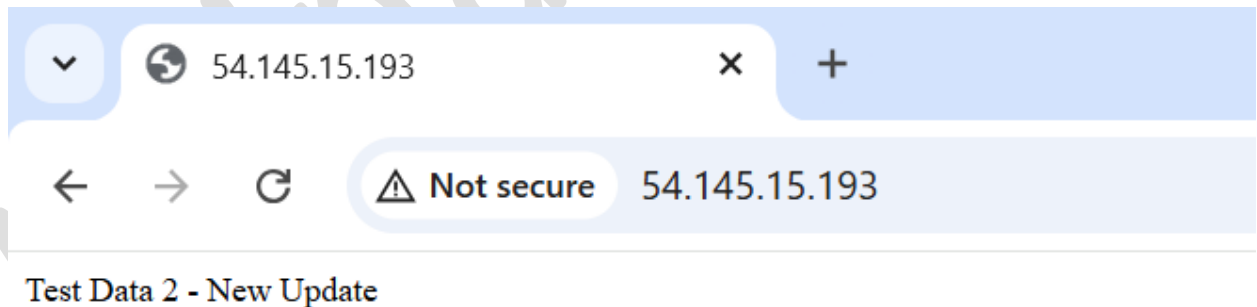
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- Refresh the **frontend**; you should now see the message content.



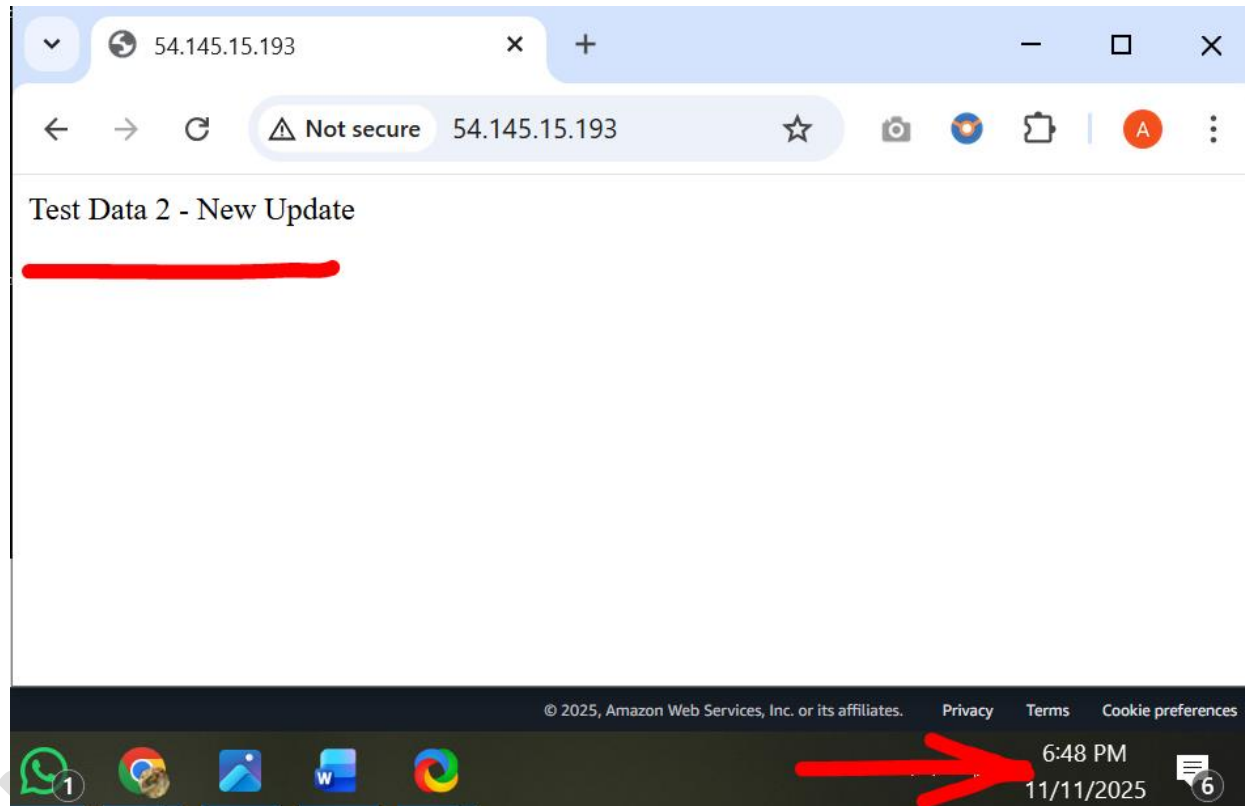
`docker exec backend sh -c "echo 'Test Data 2 - New Update' > /data/message.txt"`



8. **Stop and remove** the backend container, then refresh the frontend.
 - Confirm the **data is still available** (persisted in the volume).


```
ubuntu@ip-172-31-19-104:~/two-tier-app$ docker stop e1b
e1b
ubuntu@ip-172-31-19-104:~/two-tier-app$ docker rm e1b
e1b
ubuntu@ip-172-31-19-104:~/two-tier-app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED     STATUS    PORTS                               NAMES
88d7d93e5452   frontend-app   "docker-entrypoint.s..." 36 minutes ago    Up 36 minutes    0.0.0.0:80->80/tcp, [::]:80->80/tcp    frontend
ubuntu@ip-172-31-19-104:~/two-tier-app$
```

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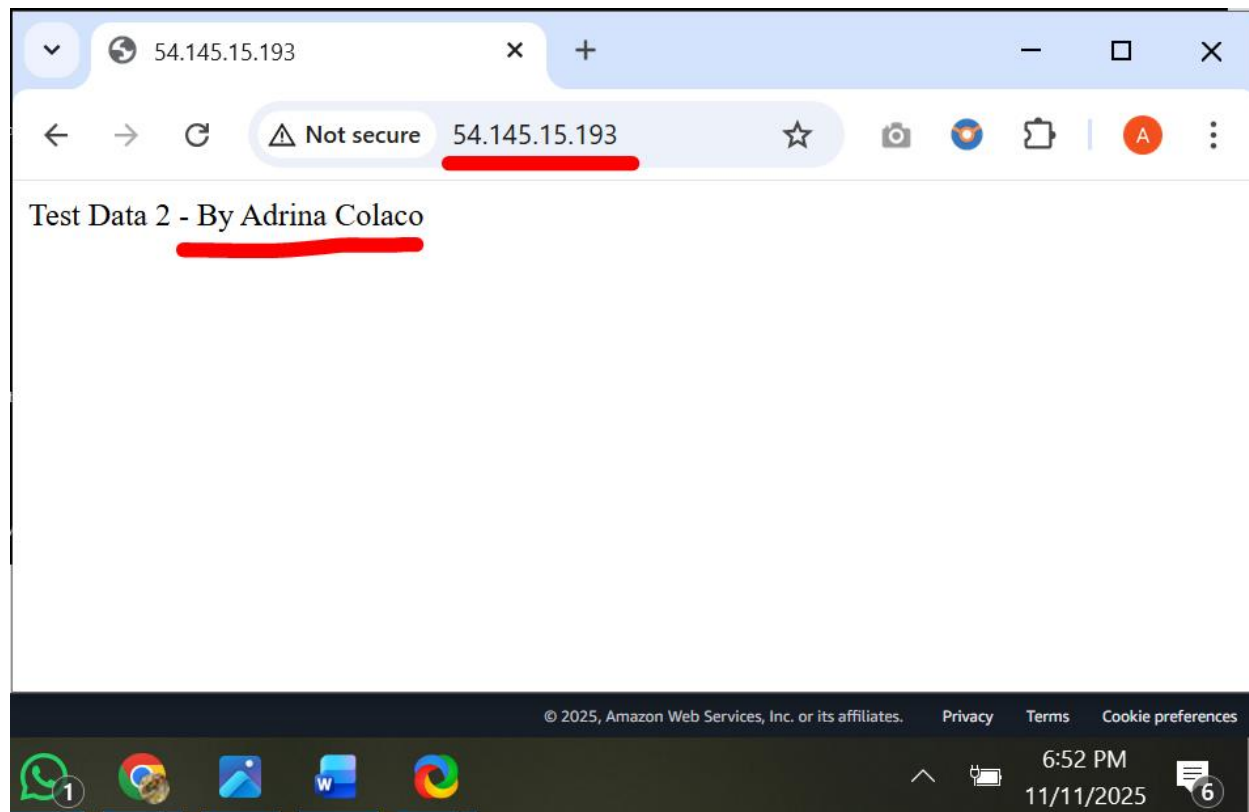


9. Re-create the backend and **write a new message**.

- Confirm the frontend now shows the **updated** content.

```
ubuntu@ip-172-31-19-104:~/two-tier-app$ docker run -d --name backend --network mynetwork -v shared-data:/data backend-app
c142fef7b3f928a3ff4833d21e10d46e853a7c04fd9d95185cfe28127d129fb7
ubuntu@ip-172-31-19-104:~/two-tier-app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED     STATUS    PORTS                               NAMES
c142fef7b3f9   backend-app    "docker-entrypoint.s..." 5 seconds ago    Up 4 seconds    80/tcp                             backend
88d7d93e5452   frontend-app   "docker-entrypoint.s..." 40 minutes ago    Up 40 minutes    0.0.0.0:80->80/tcp, [::]:80->80/tcp    frontend
ubuntu@ip-172-31-19-104:~/two-tier-app$ docker exec backend sh -c "echo 'Test Data 2 - By Adrina Colaco' > /data/message.txt"
ubuntu@ip-172-31-19-104:~/two-tier-app$
```

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Assignment 47: Docker Compose — One Command, Three Tiers (DB + API + UI)

1. Stop all containers & Remove all stopped containers

```
ubuntu@ip-172-31-19-104:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS
c142fef7b3f9   backend-app    "docker-entrypoint.s..." 18 hours ago Up 18 hours 80/tcp
88d7d93e5452   frontend-app   "docker-entrypoint.s..." 18 hours ago Up 18 hours 0.0.0.0:80->80/tcp, [::]:80->80/tcp
ubuntu@ip-172-31-19-104:~$ docker stop c14 88d
c14
88d
ubuntu@ip-172-31-19-104:~$ docker rm c14 88d
c14
88d
ubuntu@ip-172-31-19-104:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS    NAMES
ubuntu@ip-172-31-19-104:~$
```

2. Remove All Docker Images

After stopping and removing containers, remove all images to free up space.

- a. List All Images
docker images
- b. Remove All Images
docker rmi -f \$(docker images -q)

```

ubuntu@ip-172-31-19-104:~$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
frontend-app        latest         7db4ca30f26c   18 hours ago   1.1GB
backend-app         latest         cf6487b5f8c0   18 hours ago   1.1GB
node                 18            b50082bc3670   7 months ago   1.09GB
ubuntu@ip-172-31-19-104:~$ docker rmi frontend-app:latest backend-app:latest node:18
Untagged: frontend-app:latest
Deleted: sha256:7db4ca30f26c62e0f26a2b5d4d30c330a74fc935b770f699ca5c2ddc189100a3
Deleted: sha256:1ccb7faa2829d66610f84126b6a9ce66ff0ce5ea6915df986dfc0af522865fa8
Deleted: sha256:0ba9282c8fcc4b5ece317dac97d8e91f0b735f9a304f48933cc51361549b7e51
Deleted: sha256:553f99dc45588cb778b485f905b5fba6a063c5b711fcb92ab08f17b4396cc158
Untagged: backend-app:latest
Deleted: sha256:cf6487b5f8c048e34aeb47a5ce16940344e265545e1ae817147a18346cf4091
Deleted: sha256:0aa59800d3cc65a848bbeb2778a701133b35c52623578cb8a60f972b9c4f9679
Deleted: sha256:056ab6ef04aeefe7475c1946fb79c63b92106d04b1ac6b3d23dc3ca5f117607f
Deleted: sha256:4380fa1f1826a076205c0e77beea2d40fe4a3251425646d56f44a979aba22223
Deleted: sha256:a71f5fade9da5f0b2d6bcc95955539c8e7ac843da1b1fa51f7ee463a30a8c4bd
Deleted: sha256:887b3733b3eff579bc931f91ae6a73bf6d699cf4dcf1cab17a169b415fba4155
Deleted: sha256:448715c931427b676363f14070ea1c6737fd9a66ca81d4ca6e2f3817c369413e
Deleted: sha256:9f592f4f3b34b1fea4672a79558176426c9be19f10dacc32985d922095e97cbe
Untagged: node:18
Untagged: node@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
Deleted: sha256:b50082bc3670d0396b2d90e4b0e5bb10265ba5d0ee16bf40f9a505f7045ee563
Deleted: sha256:9399953f05c0204ac753b62a8b2e12cc2f90b436cf85d8bb67c348e9bc857d31
Deleted: sha256:3b39a52acf6d9ba12d12c648e2824ded532531d1667f517431cc9fbc13a81eea
Deleted: sha256:706c4917a969d92665fec82a8eb1c432d882c1e8cbeeb65943db1753b6d0182b
Deleted: sha256:f46c4f784bb5c653848d9f35c5a121837142df47b1f558f3fced70b3de409dc0
Deleted: sha256:e298f1c7405e6a8f5911d8a020c7f2551f9c0c92075a2a7d4fbf5d37300f5333
Deleted: sha256:79d1035c57159ced90a03814388505a0688a69c231e7411754c443caa2856434
Deleted: sha256:3e41611129669e647d0f2d37d15935e8f85aa73a75f8d30487f403e3e3bc5120
Deleted: sha256:2f7436e79a0bc6cdec6536da54ae6a5863c8e3b5f145e0f8ac0b96306baddbc9
ubuntu@ip-172-31-19-104:~$ docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
ubuntu@ip-172-31-19-104:~$

```

i-09882c3c7cbf4144b (DockerDevOps)

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3. Remove All Unused Docker Data (Volumes, Networks, and Build Cache)

To completely clean up the system, remove unused volumes, networks, and build cache.

- Remove Unused Volumes
docker volume prune -f
- Remove Unused Networks
docker network prune -f
- Remove Build Cache (Dangling Images, Unused Layers)
docker system prune -a -f

This command removes all stopped containers, unused networks, dangling images, and build cache.

```

ubuntu@ip-172-31-19-104:~$ docker volume prune -f
Total reclaimed space: 0B
ubuntu@ip-172-31-19-104:~$ docker network prune -f
Deleted Networks:
mynetwork

ubuntu@ip-172-31-19-104:~$ docker system prune -a -f
Total reclaimed space: 0B
ubuntu@ip-172-31-19-104:~$

```

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4. Final Verification

Run the following commands to confirm everything is removed:

Check that no containers are running or stopped:

```
docker ps -a
```

Check that no images exist:

```
docker images
```

Check that no volumes exist:

```
docker volume ls
```

```

ubuntu@ip-172-31-19-104:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-19-104:~$ docker images
REPOSITORY    TAG       IMAGE ID   CREATED   SIZE
ubuntu@ip-172-31-19-104:~$ docker volume ls
DRIVER        VOLUME NAME
ubuntu@ip-172-31-19-104:~$

```

i-09882c3c7cbf4144b (DockerDevOps)

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5. Define the Project Structure

Create the following folder structure:

```
mkdir -p multi-tier-app/{frontend,backend,database}
```

```
cd multi-tier-app
```

```
touch docker-compose.yml
```

```
ubuntu@ip-172-31-19-104:~$ mkdir -p multi-tier-app/{frontend,backend,database}
cd multi-tier-app
touch docker-compose.yml
ubuntu@ip-172-31-19-104:~/multi-tier-app$
```

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6. Final structure:

```
multi-tier-app/
| -- frontend/
|   |____ Dockerfile
| -- backend/
|   |____ Dockerfile
| -- database/
|   |____ Dockerfile
| -- docker-compose.yml
```

7. Write the docker-compose.yml File

Inside multi-tier-app/docker-compose.yml, add the following configuration:

```
version: "3.8"
```

```
services:
```

```
  database:
```

```
    build: ./database
```

```
    container_name: db
```

```
    restart: always
```

```
    networks:
```

```
      - backend-network
```

```
    volumes:
```

```
      - mongo-data:/data/db
```

```
    ports:
```

```
      - "27017:27017"
```

```
  backend:
```

```
build: ./backend
container_name: api
restart: always
depends_on:
  - database
networks:
  - backend-network
  - frontend-network
```

```
frontend:
  build: ./frontend
  container_name: ui
  restart: always
  depends_on:
    - backend
  networks:
    - frontend-network
  ports:
    - "80:80"
```

```
networks:
  backend-network:
  frontend-network:
```

```
volumes:
  mongo-data:
```

```
GNU nano 7.2 docker-compose.yml
version: "3.8"

services:
  database:
    build: ./database
    container_name: db
    restart: always
    networks:
      - backend-network
    volumes:
      - mongo-data:/data/db
    ports:
      - "27017:27017"

  backend:
    build: ./backend
    container_name: api
    restart: always
    depends_on:
      - database
    networks:
      - backend-network
      - frontend-network

  frontend:
    build: ./frontend
    container_name: ui
    restart: always
    depends_on:
      - backend
    networks:
      - frontend-network
    ports:
      - "80:80"

networks:
  backend-network:
  frontend-network:

volumes:
  mongo-data:
```

8. Create the Dockerfiles

Database (MongoDB)

Inside multi-tier-app/database/Dockerfile:

```
# Use the official MongoDB image
```

```
FROM mongo:latest
```

```
# Expose MongoDB's default port
```

EXPOSE 27017

```
# Use the official MongoDB image
FROM mongo:latest

# Expose MongoDB's default port
EXPOSE 27017
```

```
ading... ]
^G Helpes ]      ^O Write Out      ^W Where Is
] To Bracket    M-Q Previous    ^B Back
◀ Prev Word      ^A Home
^X ExitLine      M^R Read File    ^\ Replaceck
Where Was        M-W Next
F Forward
▶ Next Word      ^E End
N Next Line      M-▼ Scroll Down    ^▼ Next Block
```

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9. Backend (Node.js with Express)
Inside multi-tier-app/backend/Dockerfile:

```
# Use the official Node.js image
FROM node:18
```

```
# Set working directory
WORKDIR /app
```


Copy package.json and install dependencies

COPY package*.json ./

RUN npm init -y && npm install express mongoose cors body-parser

Copy application code

COPY . .

Create index.js with Express API and MongoDB connection

RUN echo "const express = require('express'); \

const mongoose = require('mongoose'); \

const app = express(); \

const PORT = 80; \

mongoose.connect('mongodb://db:27017/mydb', { useNewUrlParser: true, useUnifiedTopology: true }) \

.then(() => console.log('Connected to MongoDB')) \

.catch(err => console.log('MongoDB connection error:', err)); \

app.get('/', (req, res) => res.send('Hello from Backend')); \

app.listen(PORT, () => console.log('Backend running on port ' + PORT));" > index.js

Expose port 80

EXPOSE 80

Run the application

CMD ["node", "index.js"]

```
GNU nano 7.2 Dockerfile
# Use the official Node.js image
FROM node:18

# Set working directory
WORKDIR /app

# Copy package.json and install dependencies
RUN npm init -y && npm install express mongoose cors body-parser

# Copy application code
COPY . .

# Create index.js with Express API and MongoDB connection
RUN echo "const express = require('express'); \
const mongoose = require('mongoose'); \
const app = express(); \
const PORT = 80; \
mongoose.connect('mongodb://db:27017/mydb', { useNewUrlParser: true, useUnifiedTopology: true }) \
.then(() => console.log('Connected to MongoDB')) \
.catch(err => console.log('MongoDB connection error:', err)); \
app.get('/', (req, res) => res.send('Hello from Backend')); \
app.listen(PORT, () => console.log('Backend running on port ' + PORT));" > index.js

# Expose port 80
EXPOSE 80

# Run the application
CMD ["node", "index.js"]

[ Wrote 28 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line  M-E Redo
```

i-09882c3c7cbf4144b (DockerDevOps)

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10. Frontend (React with Nginx)

Inside multi-tier-app/frontend/Dockerfile:

```
# Use the official lightweight Nginx image
```

```
FROM nginx:alpine
```

```
# Expose port 80 for incoming traffic
```

```
EXPOSE 80
```

```
# Start Nginx in the foreground
```

```
CMD ["nginx", "-g", "daemon off;"]
```

```
# Use the official lightweight Nginx image
FROM nginx:alpine

# Expose port 80 for incoming traffic
EXPOSE 80

# Start Nginx in the foreground
CMD ["nginx", "-g", "daemon off;"]
```

11. Build and Start the Application

Install docker-compose

sudo apt install docker-compose

[illegible]

Navigate to multi-tier-app/ and run:
docker-compose up --build -d
--build forces a rebuild of images.
-d runs services in the background.

Adriana Colaco

12. Verify Everything is Running

Check Running Containers

`docker ps`

Expected output:

CONTAINER ID	IMAGE	STATUS	NAMES
a1b2c3d4e5f6	frontend-app	Up 10 min	ui
b2c3d4e5f6a1	backend-app	Up 10 min	api
c3d4e5f6a1b2	mongo	Up 10 min	db

```
ubuntu@ip-172-31-19-104:~/multi-tier-app$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
e353f70a4cd8   multi-tier-app_frontend             "/docker-entrypoint.s..." About a minute ago Up About a minute   0.0.0.0:80->80/tcp, [::]:80->80/tcp   ui
7e2bac83ef39   multi-tier-app_backend              "docker-entrypoint.s..." About a minute ago Up About a minute   80/tcp                               api
fda9bef67715   multi-tier-app_database              "docker-entrypoint.s..." About a minute ago Up About a minute   0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp   db
ubuntu@ip-172-31-19-104:~/multi-tier-app$
```

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13. Test the Application

Check Frontend (React)

Since the frontend is exposed on port 80, test it:

`curl http://localhost`

Expected output:

Welcome to nginx!

```
ubuntu@ip-172-31-19-104:~/multi-tier-app$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
ubuntu@ip-172-31-19-104:~/multi-tier-app$
```

i-09882c3c7cbf4144b (DockerDevOps)

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Check Backend API

Since the backend is also exposed on port 80, test it:

curl http://localhost

Expected output:

Hello from Backend

```
ubuntu@ip-172-31-19-104:~/multi-tier-app$ docker exec -it ui /bin/sh
/ # curl http://backend
Hello from Backend/ #
```

i-09882c3c7cbf4144b (DockerDevOps)

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14. Manage Services

Restart the Backend

docker-compose restart backend

Stop the Entire Application

docker-compose down


```

ubuntu@ip-172-31-19-104:~/multi-tier-app$ docker-compose restart backend
Restarting api ... done
ubuntu@ip-172-31-19-104:~/multi-tier-app$ docker-compose down
Stopping ui ... done
Stopping api ... done
Stopping db ... done
Removing ui ... done
Removing api ... done
Removing db ... done
Removing network multi-tier-app_backend-network
Removing network multi-tier-app_frontend-network
ubuntu@ip-172-31-19-104:~/multi-tier-app$

```

i-09882c3c7cbf4144b (DockerDevOps)

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Assignment 48: Deploy the Book Review App with Docker Compose (MySQL + API + UI)

1. Clone the Repository

git clone https://github.com/pravinmishraaws/book-review-app.git

cd book-review-app

```

ubuntu@ip-172-31-73-168:~$ git clone https://github.com/pravinmishraaws/book-review-app.git
cd book-review-app
Cloning into 'book-review-app'...
remote: Enumerating objects: 4049, done.
remote: Counting objects: 100% (61/61), done.
remote: Compressing objects: 100% (17/17), done.
remote: Total 4049 (delta 51), reused 44 (delta 44), pack-reused 3988 (from 2)
Receiving objects: 100% (4049/4049), 4.27 MiB | 11.01 MiB/s, done.
Resolving deltas: 100% (1004/1004), done.
ubuntu@ip-172-31-73-168:~/book-review-app$

```

i-0acdf37be9a831dfc (DockerDevOps)

PublicIPs: 44.221.49.162 PrivateIPs: 172.31.73.168

2. Add .dockerignore to Frontend and Backend

Create files to keep images lean:

frontend/.dockerignore

node_modules

build

Dockerfile

.dockerignore

```
ubuntu@ip-172-31-19-104:~/book-review-app$ cd frontend/  
ubuntu@ip-172-31-19-104:~/book-review-app/frontend$ touch .dockerignore  
ubuntu@ip-172-31-19-104:~/book-review-app/frontend$ nano .dockerignore  
ubuntu@ip-172-31-19-104:~/book-review-app/frontend$
```

```
GNU nano 7.2 .dockerignore *  
node_modules  
build  
Dockerfile  
.dockerignore  
|
```

backend/.dockerignore

node_modules
Dockerfile
.dockerignore

```
ubuntu@ip-172-31-19-104:~/book-review-app$ cd backend/  
ubuntu@ip-172-31-19-104:~/book-review-app/backend$ nano .dockerignore  
ubuntu@ip-172-31-19-104:~/book-review-app/backend$
```

```
GNU nano 7.2 .dockerignore  
node_modules  
Dockerfile  
.dockerignore  
|
```

3. Verify Dockerfiles

Confirm the repo already has frontend/Dockerfile and backend/Dockerfile.
If missing, create them as below:

backend/Dockerfile

```
FROM node:18  
WORKDIR /app  
COPY package*.json ./  
RUN npm install  
COPY . .  
EXPOSE 3001  
CMD ["node", "src/server.js"]
```

```

GNU nano 7.2 Dockerfile
# Use Node.js LTS as base image
FROM node:18

# Set working directory
WORKDIR /app

# Copy package.json and package-lock.json
COPY package*.json ./

# Install dependencies
RUN npm install

# Copy the rest of the application files
COPY . .

# Expose the port the backend runs on
EXPOSE 3001

# Start the backend server
#CMD ["node", "src/server.js"]
CMD ["node", "src/server.js"]

```

frontend/Dockerfile

```

FROM node:18
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
CMD ["npm", "run", "dev"]

```

```

GNU nano 7.2 Dockerfile
# Use Node.js LTS as base image
FROM node:18

# Set working directory
WORKDIR /app

# Copy package.json and package-lock.json
COPY package*.json ./

# Install dependencies
RUN npm install

# Copy the rest of the application files
COPY . .

# Expose the Next.js port
EXPOSE 3000

# Start Next.js in development mode
CMD ["npm", "run", "dev"]

```

4. Create docker-compose.yml
At the project root (book-review-app/), create:

version: "3.8"

services:

mysql:

image: mysql:8

container_name: mysql

restart: always

environment:

MYSQL_ROOT_PASSWORD: root123

MYSQL_DATABASE: bookdb

MYSQL_USER: bookuser

MYSQL_PASSWORD: bookpass

volumes:

- mysql_data:/var/lib/mysql

healthcheck:

test: ["CMD", "mysqladmin", "ping", "-h", "localhost", "-uroot", "-proot123"]

interval: 10s

timeout: 5s

retries: 5

backend:

build: ./backend

container_name: backend

restart: always

depends_on:

mysql:

condition: service_healthy

environment:

DB_HOST: mysql

DB_USER: bookuser

DB_PASSWORD: bookpass

DB_NAME: bookdb

DB_PORT: 3306

JWT_SECRET: mysecret

CORS_ORIGIN: http://<YOUR_VM_PUBLIC_IP>:3000

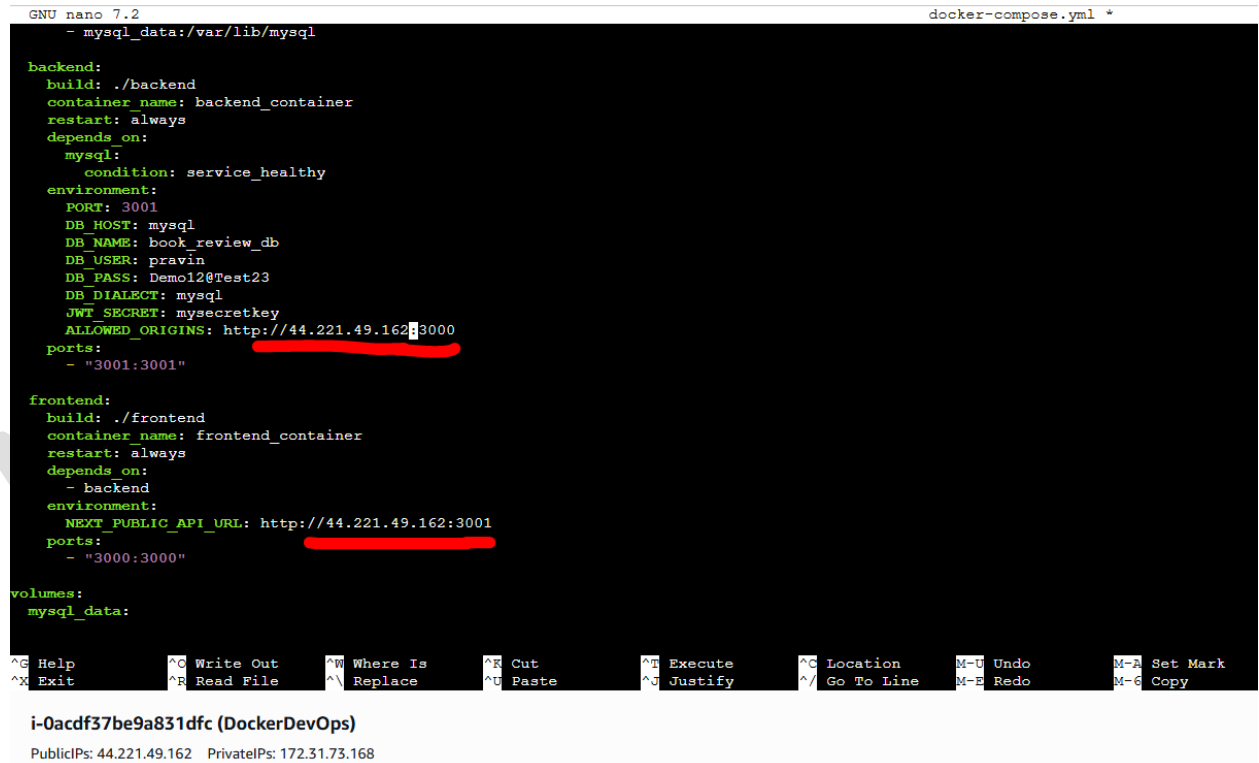
ports:

- "3001:3001"

```
frontend:
  build: ./frontend
  container_name: frontend
  restart: always
  depends_on:
    - backend
  environment:
    REACT_APP_API_URL: http://backend:3001
  ports:
    - "3000:3000"
```

```
volumes:
  mysql_data:
```

◆ Replace <YOUR_VM_PUBLIC_IP> with your actual EC2 public IP.



```
GNU nano 7.2 docker-compose.yml *
- mysql_data:/var/lib/mysql

backend:
  build: ./backend
  container_name: backend_container
  restart: always
  depends_on:
    mysql:
      condition: service_healthy
  environment:
    PORT: 3001
    DB_HOST: mysql
    DB_NAME: book_review_db
    DB_USER: pravin
    DB_PASS: Demo12@Test23
    DB_DIALECT: mysql
    JWT_SECRET: mysecretkey
    ALLOWED_ORIGINS: http://44.221.49.162:3000
  ports:
    - "3001:3001"

frontend:
  build: ./frontend
  container_name: frontend_container
  restart: always
  depends_on:
    - backend
  environment:
    NEXT_PUBLIC_API_URL: http://44.221.49.162:3001
  ports:
    - "3000:3000"

volumes:
  mysql_data:
```

i-0acdf37be9a831dfc (DockerDevOps)

PublicIPs: 44.221.49.162 PrivateIPs: 172.31.73.168

5. Start the Stack

Run:

```
docker-compose up -d
```

```
shanto@ip-172-31-73-168:~/hook-review-app$ docker-compose up -d
Pulling mysql (mysql:8.0)...
023e182e2a5f: Pull complete
4f78e34adfad: Pull complete
3e4d1982d9e2: Pull complete
c5ecfb07ed08: Pull complete
4f98ea4123bf: Pull complete
2a2d32544031: Pull complete
48ec49971d94: Pull complete
f2da9f528d44: Pull complete
abc3f002a6d9: Pull complete
37bd5164f763: Pull complete
469710e4f4e9: Pull complete
Digest: sha256:c37951fc3753a6a22d6c7bf6978c5e5f6cf6f31814d98c502524f98ea52b21
Status: Downloaded newer image for mysql:8.0
Building backend
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  89.6kB
Step 1/7 : FROM node:18
18: Pulling from library/node
3e4b3d1a9511: Pulling fs layer
37927ed901b1: Pulling fs layer
78a2f47ad444: Pulling fs layer
a23f099911d6: Pulling fs layer
ada7f44f2b0d: Pulling fs layer
c4b30c3f1696: Pulling fs layer
3697bc50c9b8: Pulling fs layer
461077a72fb7: Pulling fs layer
a23f099911d6: Waiting
ada7f44f2b0d: Waiting
c4b30c3f1696: Waiting
3697bc50c9b8: Waiting
461077a72fb7: Waiting
37927ed901b1: Verifying Checksum
37927ed901b1: Download complete
3e4b3d1a9511: Verifying Checksum
3e4b3d1a9511: Download complete
ada7f44f2b0d: Verifying Checksum
ada7f44f2b0d: Download complete
78a2f47ad444: Verifying Checksum
78a2f47ad444: Download complete
3697bc50c9b8: Verifying Checksum
3697bc50c9b8: Download complete
461077a72fb7: Verifying Checksum
461077a72fb7: Download complete
c4b30c3f1696: Verifying Checksum
c4b30c3f1696: Download complete
a23f099911d6: Verifying Checksum
a23f099911d6: Download complete
3e4b3d1a9511: Pull complete
37927ed901b1: Pull complete
78a2f47ad444: Pull complete
a23f099911d6: Pull complete
ada7f44f2b0d: Pull complete
c4b30c3f1696: Pull complete
3697bc50c9b8: Pull complete
461077a72fb7: Pull complete
Digest: sha256:6da679e38498325db67193d1916ec1d224d96c693a84d943498556716d3783
Status: Downloaded newer image for node:18
--> b50082bc3670
Step 2/7 : WORKDIR /app
--> Running in 2421fc6f67b
--> Removed intermediate container 2421fc6f67b
--> d67475c2bc71
Step 3/7 : COPY package*.json ./
--> 27ad819ac59
Step 4/7 : RUN npm install
--> Running in cf9fd5e3bf0c
added 119 packages, and audited 120 packages in 3s
11 packages are looking for funding
  run `npm fund` for details
1 moderate severity vulnerability
To address all issues, run:
  npm audit fix
Run `npm audit` for details.
npm notice New major version of npm available! 10.8.2 -> 11.6.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.2
npm notice To update run: npm install -g npm@11.6.2
npm notice
--> Removed intermediate container cf9fd5e3bf0c
--> 16d478f8c214
Step 5/7 : COPY . .
--> 3984d43c2a1
Step 6/7 : EXPOSE 3001
--> Running in 5147702b633b
--> Removed intermediate container 5147702b633b
--> 5e4f8e3e27dfc
Step 7/7 : CMD ["node", "src/server.js"]
--> Running in a9e9e07d3bd4
--> Removed intermediate container a9e9e07d3bd4
--> a18f2e1c157
Successfully built a18f2e1c157
Successfully tagged hook-review-app:backend:latest
WARNING: Image for service backend was built because it did not already exist. To rebuild this image you must use `docker-compose build` or `docker-compose up --build`.
Building frontend
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  248.8kB
Step 1/7 : FROM node:18
--> b50082bc3670
Step 2/7 : WORKDIR /app
--> Using cache
--> d67475c2bc71
Step 3/7 : COPY package*.json ./
--> 81ec0ba1ab2
Step 4/7 : RUN npm install
--> Running in 91efc934964d
added 326 packages, and audited 327 packages in 22s
135 packages are looking for funding
  run `npm fund` for details
6 vulnerabilities (3 low, 1 moderate, 1 high, 1 critical)
To address issues that do not require attention, run:
  npm audit fix
To address all issues, run:
  npm audit fix --force
Run `npm audit` for details.
npm notice New major version of npm available! 10.8.2 -> 11.6.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.2
npm notice To update run: npm install -g npm@11.6.2
npm notice
--> Removed intermediate container 91efc934964d
--> c74d18e1659
Step 5/7 : COPY . .
--> bdf7ba73e39
Step 6/7 : EXPOSE 3000
--> Running in 0f10642830d3
--> Removed intermediate container 0f10642830d3
--> 1980542803d4
Step 7/7 : CMD ["npm", "run", "dev"]
--> Running in 5827a013dd4d
--> Removed intermediate container 5827a013dd4d
--> a1d42e637ac4
Successfully built a1d42e637ac4
Successfully tagged hook-review-app:frontend:latest
WARNING: Image for service frontend was built because it did not already exist. To rebuild this image you must use `docker-compose build` or `docker-compose up --build`.
Creating mysql container ... done
Creating backend container ... done
Creating frontend container ... done
shanto@ip-172-31-73-168:~/hook-review-app$
```

l-QacdF37be9a831dfc (DockerDevOps)

Public IP: 44.221.49.162 Private IP: 172.31.73.168

Then check:
docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
4d06bc7bd5a1	book-review-app_frontend	"docker-entrypoint.s..."	46 seconds ago	Up 45 seconds	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	frontend_container
12675c277de0	book-review-app_backend	"docker-entrypoint.s..."	46 seconds ago	Up 45 seconds	0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp	backend_container
9f92d0404f8c	mysql:8.0	"docker-entrypoint.s..."	57 seconds ago	Up 56 seconds (healthy)	0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp	mysql_container

i-0acdf37be9a831dfc (DockerDevOps)
PublicIPs: 44.221.49.162 PrivateIPs: 172.31.73.168

View logs if needed:
docker-compose logs -f

aws

Q Search

[Alt+S]

United States (N. Virginia)

Account ID: 3111-4154-2113

Adrina Colaco

```
ubuntu@ip-172-31-73-168:~/book-review-app$ docker-compose logs -f
Attaching to frontend_container, backend_container, mysql_container
backend_container | Database 'book_review_db' connected successfully with SSL!
backend_container | Database schema updated successfully!
backend_container | 3 Sample users added!
backend_container | 3 Sample books added!
backend_container | 3 Sample reviews added!
backend_container | Server running on port 3001
frontend_container |
frontend_container | > frontend@0.1.0 dev
frontend_container | > next dev --turbo
frontend_container |
frontend_container | ▲ Next.js 15.2.3 (TurboPack)
frontend_container |   - Local:    http://localhost:3000
frontend_container |   - Network:  http://172.18.0.4:3000
frontend_container |
frontend_container | ✓ Starting...
frontend_container | Attention: Next.js now collects completely anonymous telemetry regarding usage.
frontend_container | This information is used to shape Next.js' roadmap and prioritize features.
frontend_container | You can learn more, including how to opt-out if you'd not like to participate in this anonymous program, by visiting the following URL:
frontend_container | https://nextjs.org/telemetry
frontend_container |
frontend_container | ✓ Ready in 1732ms
mysql_container | 2025-11-13 05:06:10:00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
mysql_container | 2025-11-13 05:06:10:00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
mysql_container | 2025-11-13 05:06:10:00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.44-1.el9 started.
mysql_container | 2025-11-13 05:06:11:4930152 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
mysql_container | 2025-11-13 05:06:11:4965902 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.44) starting as process 1
mysql_container | 2025-11-13 05:06:11:5224312 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
mysql_container | 2025-11-13 05:06:12:0654572 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysql_container | 2025-11-13 05:06:12:4019822 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
mysql_container | 2025-11-13 05:06:12:4020582 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
mysql_container | 2025-11-13 05:06:12:4094262 0 [Warning] [MY-011810] [Server] Insecure configuration for '--pid-file': Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
mysql_container | 2025-11-13 05:06:12:4698592 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
mysql_container | 2025-11-13 05:06:12:4901992 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.44' socket: /var/run/mysqld/mysqld.sock port: 3306 MySQL Community Server - GPL.
```

i-0acdf37be9a831dfc (DockerDevOps)

PublicIPs: 44.221.49.162 PrivateIPs: 172.31.73.168

CloudShell

Feedback

Console Mobile App

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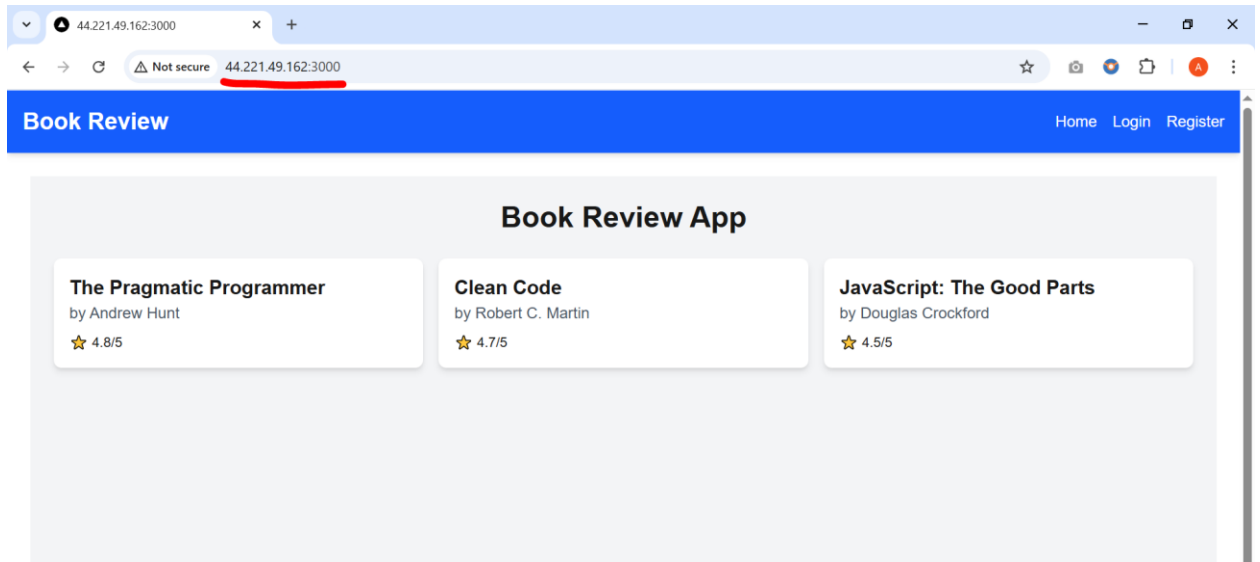
Ensure MySQL is healthy, then backend starts connecting successfully.

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
4d06bc7bd5a1	book-review-app_frontend	"docker-entrypoint.s..."	46 seconds ago	Up 45 seconds	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	frontend_container
12675c277de0	book-review-app_backend	"docker-entrypoint.s..."	46 seconds ago	Up 45 seconds	0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp	backend_container
9f92d0404f8c	mysql:8.0	"docker-entrypoint.s..."	57 seconds ago	Up 56 seconds (healthy)	0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp	mysql_container

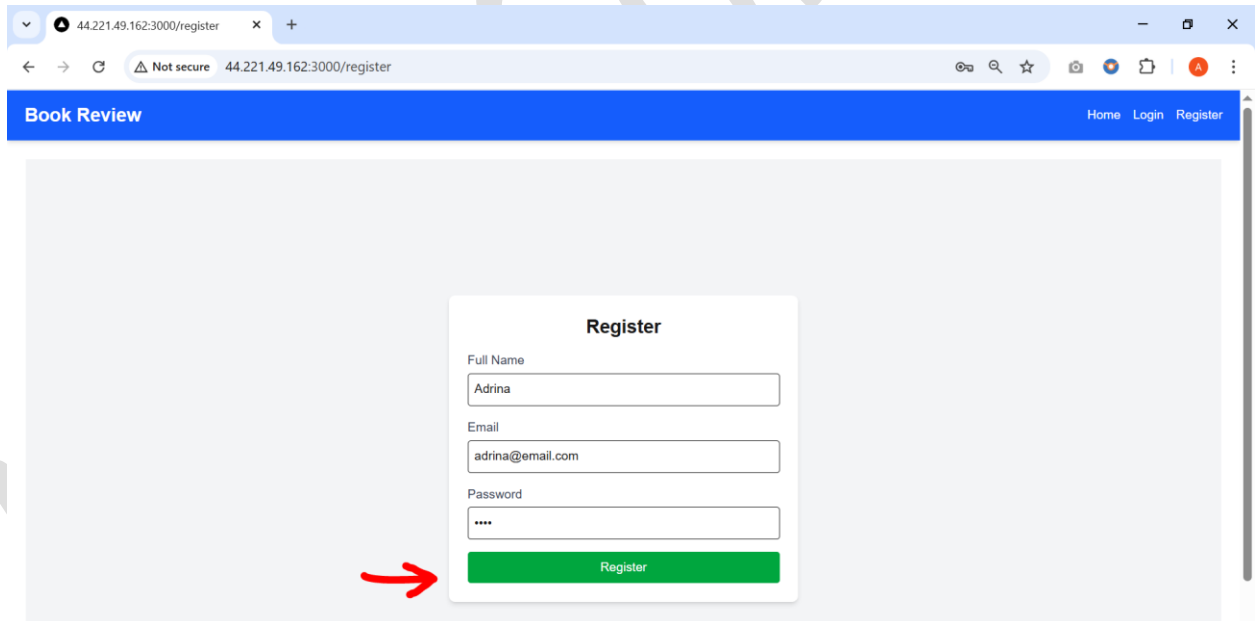
i-0acdf37be9a831dfc (DockerDevOps)
PublicIPs: 44.221.49.162 PrivateIPs: 172.31.73.168

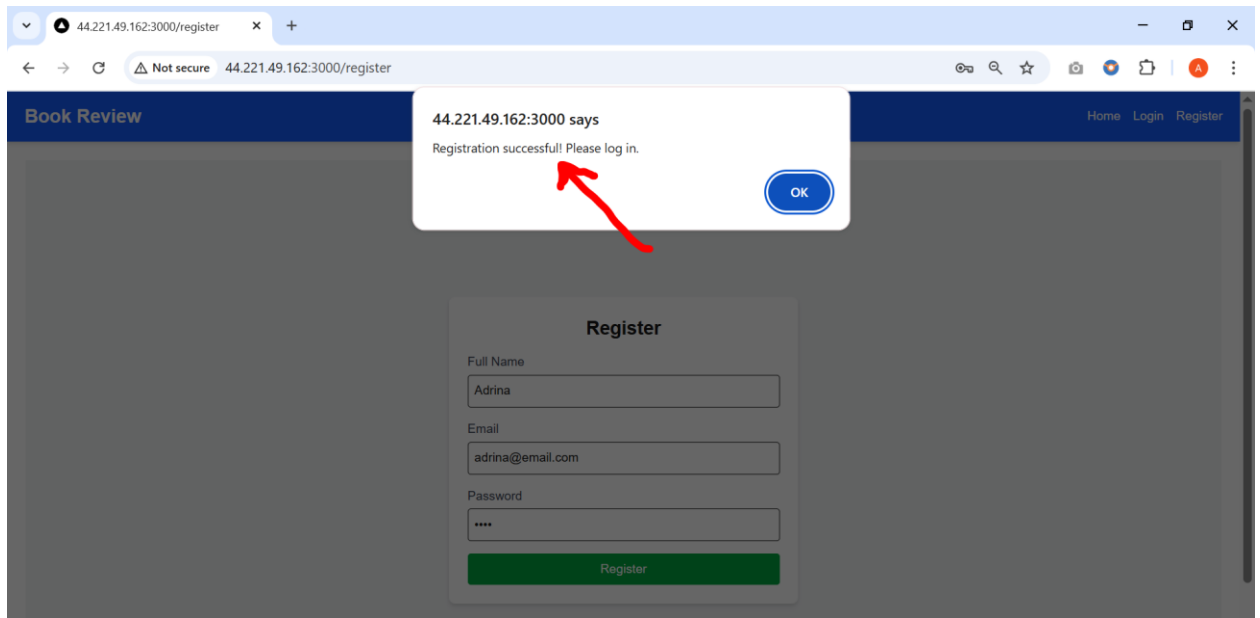
- Test in Browser
- Open your browser:

👉 http://<VM_PUBLIC_IP>:3000

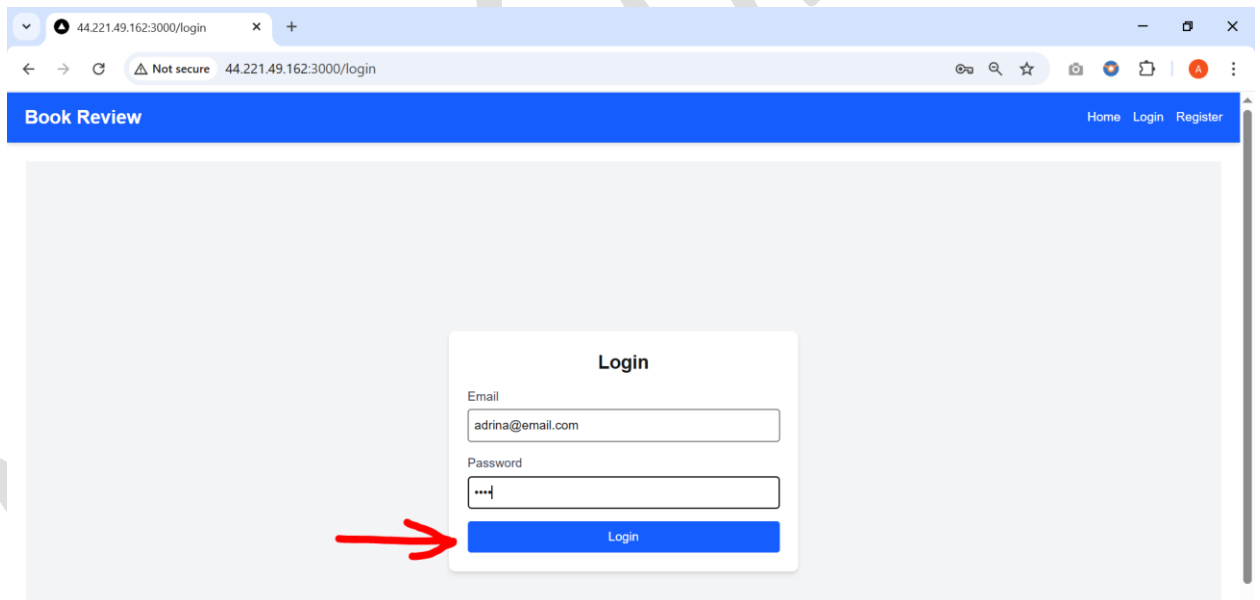


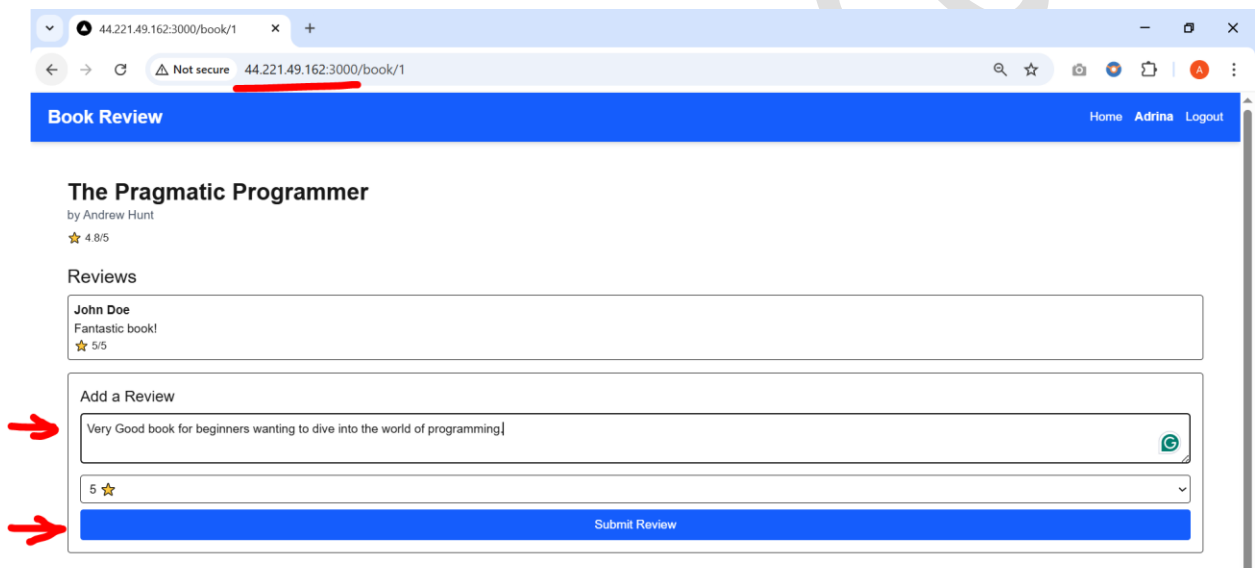
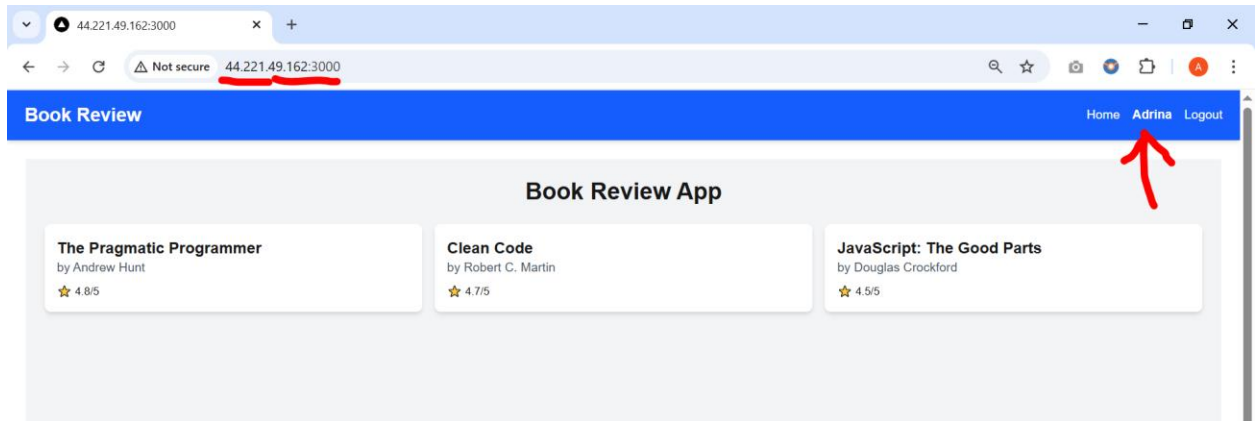
Register a new user.



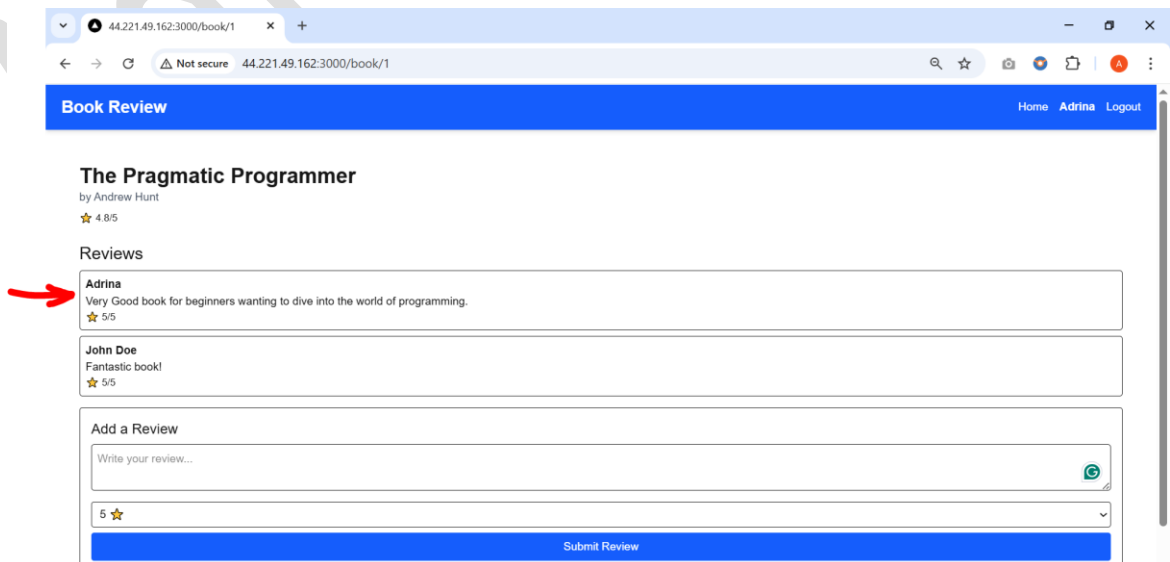


Login and add a book review.





Confirm review appears.



7. Verify Persistence

Run:

`docker-compose down`

(do not add `-v`)

```
ubuntu@ip-172-31-73-168:~/book-review-app$ docker-compose down
Stopping frontend_container ... done
Stopping backend_container ... done
Stopping mysql_container ... done
Removing frontend_container ... done
Removing backend_container ... done
Removing mysql_container ... done
Removing network book-review-app_default
ubuntu@ip-172-31-73-168:~/book-review-app$
```

i-0acdf37be9a831dfc (DockerDevOps)

PublicIPs: 44.221.49.162 PrivateIPs: 172.31.73.168

Then:

`docker-compose up -d`

```
ubuntu@ip-172-31-73-168:~/book-review-app$ docker-compose up -d
Creating network "book-review-app_default" with the default driver
Creating mysql_container ... done
Creating backend_container ... done
Creating frontend_container ... done
ubuntu@ip-172-31-73-168:~/book-review-app$
```

i-0acdf37be9a831dfc (DockerDevOps)

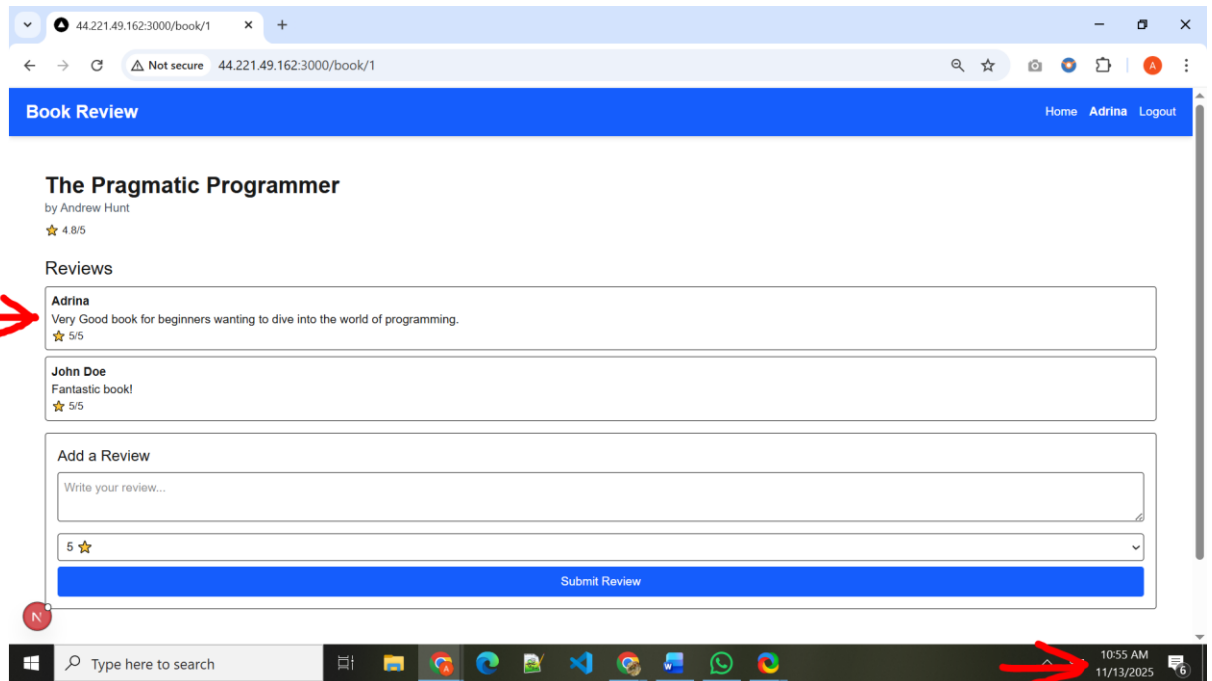
PublicIPs: 44.221.49.162 PrivateIPs: 172.31.73.168

```
ubuntu@ip-172-31-73-168:~/book-review-app$ docker-compose up -d
Creating network "book-review-app_default" with the default driver
Creating mysql_container ... done
Creating backend_container ... done
Creating frontend_container ... done
ubuntu@ip-172-31-73-168:~/book-review-app$
```

i-0acdf37be9a831dfc (DockerDevOps)
PublicIPs: 44.221.49.162 PrivateIPs: 172.31.73.168



Go back to the app — your reviews and users should still exist ☒ (stored in `mysql_data` volume).



8. Teardown (Optional)

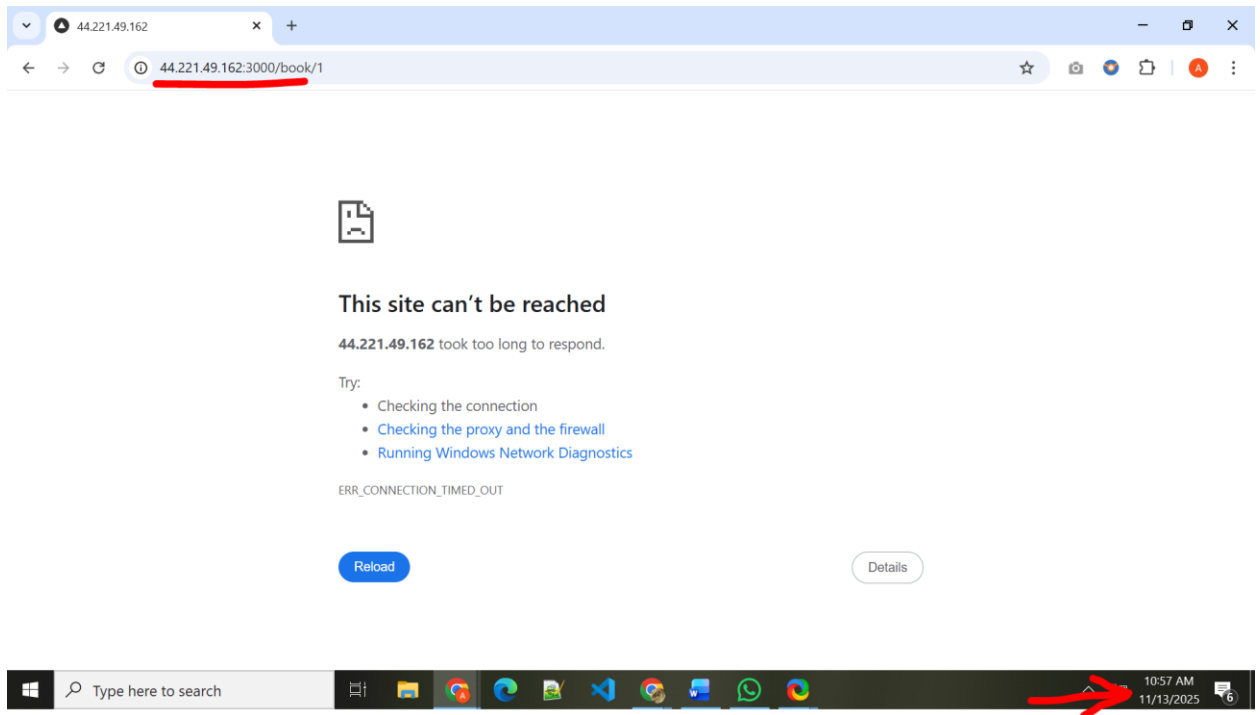
To remove everything (including volume):

`docker-compose down -v`

```
ubuntu@ip-172-31-73-168:~/book-review-app$ docker-compose down -v
Stopping frontend_container ... done
Stopping backend_container ... done
Stopping mysql_container ... done
Removing frontend_container ... done
Removing backend_container ... done
Removing mysql_container ... done
Removing network book-review-app_default
Removing volume book-review-app_mysql_data
ubuntu@ip-172-31-73-168:~/book-review-app$
```

i-0acdf37be9a831dfc (DockerDevOps)

PublicIPs: 44.221.49.162 PrivateIPs: 172.31.73.168



Assignment 49 – Capstone Project: The EpicBook — Production-Ready Dockerized Deployment

1. Clone the Repository

```
git clone https://github.com/pravinmishraaws/theepicbook.git
cd theepicbook
```

```
ubuntu@ip-172-31-68-154:~$ git clone https://github.com/pravinmishraaws/theepicbook.git
Cloning into 'theepicbook'...
remote: Enumerating objects: 156, done.
remote: Counting objects: 100% (26/26), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 156 (delta 24), reused 21 (delta 21), pack-reused 130 (from 2)
Receiving objects: 100% (156/156), 30.70 MiB | 41.92 MiB/s, done.
Resolving deltas: 100% (41/41), done.
ubuntu@ip-172-31-68-154:~$ cd theepicbook/
ubuntu@ip-172-31-68-154:~/theepicbook$
```

2. Add .dockerignore to “theepicbook” folder

Create files to keep images lean:

~/theepicbook/.dockerignore

node_modules

build

Dockerfile

.dockerignore

```
ubuntu@ip-172-31-68-154:~/theepicbook$ touch .dockerignore
ubuntu@ip-172-31-68-154:~/theepicbook$ nano .dockerignore
ubuntu@ip-172-31-68-154:~/theepicbook$
```

```
GNU nano 7.2 .dockerignore
node_modules
build
Dockerfile
.dockerignore
```

3. Verify Dockerfile

```
FROM node:18
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
#CMD ["npm", "run", "dev"]
CMD ["sh", "-c", "node src/server.js && npm run dev"]
```

```
GNU nano 7.2 Dockerfile
FROM node:18
WORKDIR /theepicbook
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
#CMD ["npm", "run", "dev"]
CMD ["sh", "-c", "node /theepicbook/server.js && npm run dev"]
```

4. Create docker-compose.yml

At the project root (book-review-app/), create:

```
version: '2.1'
```

services:

mysql:

image: mysql:8.0

container_name: mysql_container

restart: always

healthcheck:

test: ["CMD", "mysqladmin", "ping", "-h", "localhost"]

interval: 10s

timeout: 5s

retries: 5

environment:

MYSQL_ROOT_PASSWORD: my-secret-pw

MYSQL_DATABASE: bookstore

MYSQL_USER: epicdbadmin

MYSQL_PASSWORD: MyPassword_123!

ports:

- "3306:3306"

volumes:

- mysql_data:/var/lib/mysql

theepicbook:

build: ~/theepicbook

container_name: theepicbook_container

restart: always

depends_on:

mysql:

condition: service_healthy

environment:

PORT: 3000

DB_HOST: mysql

DB_NAME: bookstore

DB_USER: mysqladmin

DB_PASS: DevSecureDBPassword123!

DB_DIALECT: mysql

JWT_SECRET: mysecretkey

ALLOWED_ORIGINS: http://<< YOUR PUBLIC IP>>:3000

ports:

- "3000:3000"

volumes:

mysql_data:

◆ Replace << YOUR PUBLIC IP>> with your actual EC2 public IP.

```

version: '2.1'

services:
  mysql:
    image: mysql:8.0
    container_name: mysql_container
    restart: always
    healthcheck:
      test: ["CMD", "mysqladmin", "ping", "-h", "localhost"]
      interval: 10s
      timeout: 5s
      retries: 5
    environment:
      MYSQL_ROOT_PASSWORD: my-secret-pw
      MYSQL_DATABASE: bookstore
      MYSQL_USER: epicdbadmin
      MYSQL_PASSWORD: MyPassword_123!
    ports:
      - "3306:3306"
    volumes:
      - mysql_data:/var/lib/mysql

  theepicbook:
    build: ~/theepicbook
    container_name: theepicbook_container
    restart: always
    depends_on:
      mysql:
        condition: service_healthy
    environment:
      PORT: 3000
      DB_HOST: mysql
      DB_NAME: bookstore
      DB_USER: mysqladmin
      DB_PASS: DevSecureDBPassword123!
      DB_DIALECT: mysql
      JWT_SECRET: mysecretkey
      ALLOWED_ORIGINS: http://3.215.23.236:3000
    ports:
      - "3000:3000"

volumes:
  mysql_data:

```

5. Change Config.json to reflect the correct database credentials.

```

mysql_data:
ubuntu@ip-172-31-68-154:~/theepicbook$ cd config/
ubuntu@ip-172-31-68-154:~/theepicbook/config$ nano config.json
ubuntu@ip-172-31-68-154:~/theepicbook/config$

```

i-0ecc26f13fbaf00cf (DockerEpicbooks)

PublicIPs: 3.215.23.236 PrivateIPs: 172.31.68.154


```
GNU nano 7.2 config.json
{
  "development": {
    "username": "epicdbadmin",
    "password": "MyPassword_123!",
    "database": "bookstore",
    "host": "mysql",
    "port": "3306",
    "dialect": "mysql"
  },
  "test": {
    "username": "root",
    "password": null,
    "database": "database_test",
    "host": "127.0.0.1",
    "dialect": "mysql"
  },
  "production": {
    "use_env_variable": "JAWSDB_URL",
    "dialect": "mysql"
  }
}
```

6. Start the Stack

Run:

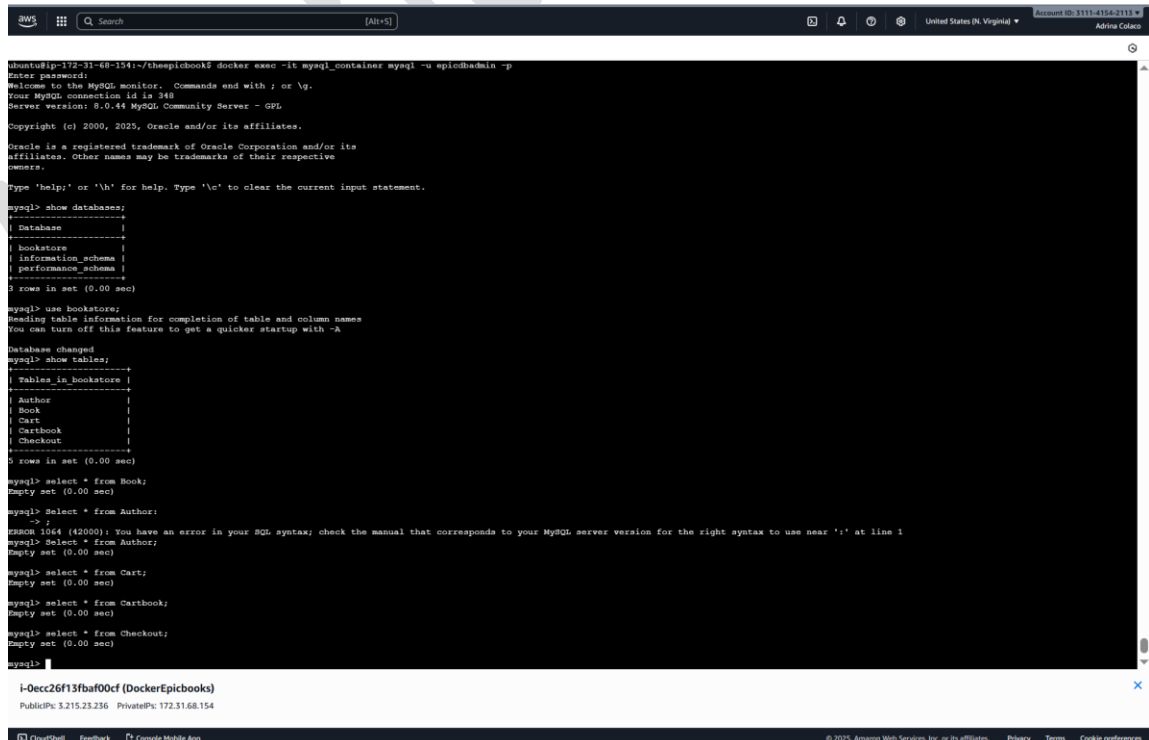
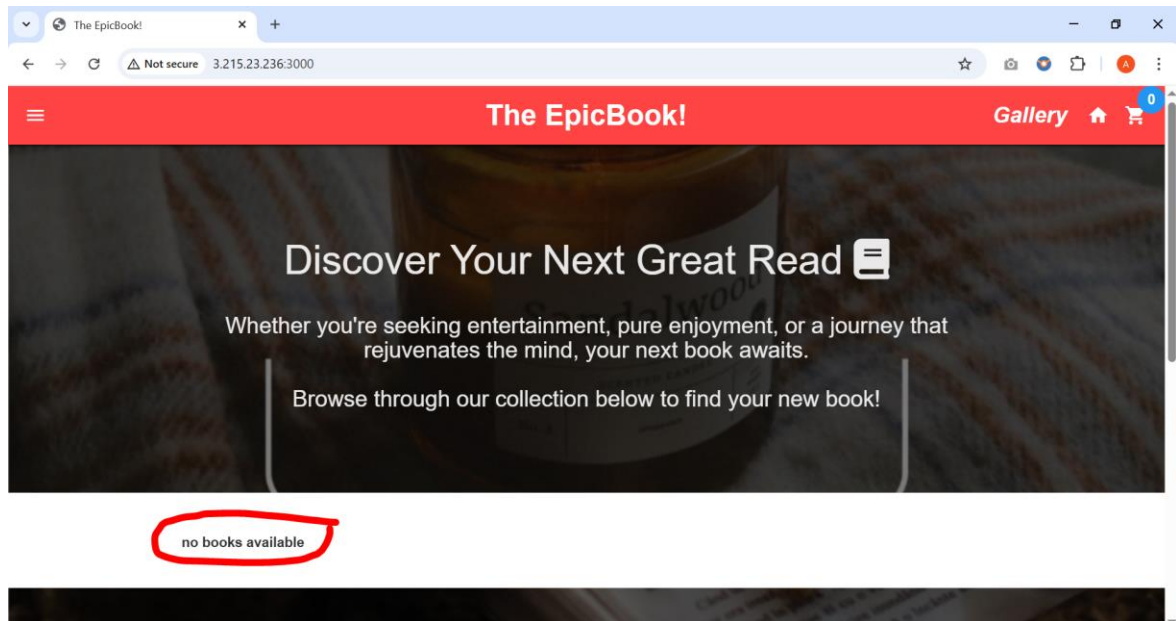
`docker-compose up -d`

Ensure MySQL is healthy, then app starts connecting successfully.

ubuntu@ip-172-31-68-154:~/theepicbook\$ docker ps									
CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES			
ad9a846376f1	theepicbook_theepicbook	"docker-entrypoint.s_"	33 minutes ago	Up 33 minutes	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	theepicbook_container			
5a6ea9c91167	mysql:8.0	"docker-entrypoint.s_"	33 minutes ago	Up 33 minutes (healthy)	0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp	mysql_container			

i-0ecc26f13fbaf00cf (DockerEpicbooks)
PublicIPs: 3.215.23.236 PrivateIPs: 172.31.68.154

7. If you run the Public IP at this point, you will not see no books on the website. That's because there is no data in the mysql tables.



8. Run the mysql scripts to add data to the database.

Since the database structure already exists, run only the SQL files that insert data into those tables.

```
ubuntu@ip-172-31-68-154:~/theepicbook/db$ docker exec -i mysql_container mysql -u epicdbadmin -pMyPassword_123! < ~/theepicbook/db/author_seed.sql
mysql: [Warning] Using a password on the command line interface can be insecure.
ubuntu@ip-172-31-68-154:~/theepicbook/db$ docker exec -i mysql_container mysql -u epicdbadmin -pMyPassword_123! < ~/theepicbook/db/books_seed.sql
mysql: [Warning] Using a password on the command line interface can be insecure.
```

i-0ecc26f13fbaf00cf (DockerEpicbooks)

PublicIPs: 3.215.23.236 PrivateIPs: 172.31.68.154

Use Docker EXEC mode to see if data is inserted in the Mysql tables

```
aws [Alt+S]

ubuntu@ip-172-31-68-154:~/theepicbook$ docker exec -it mysql_container mysql -u epicdbadmin -pMyPassword_123!
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 460
Server version: 8.0.44 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| bookstore |
| information_schema |
| performance_schema |
+-----+
3 rows in set (0.00 sec)

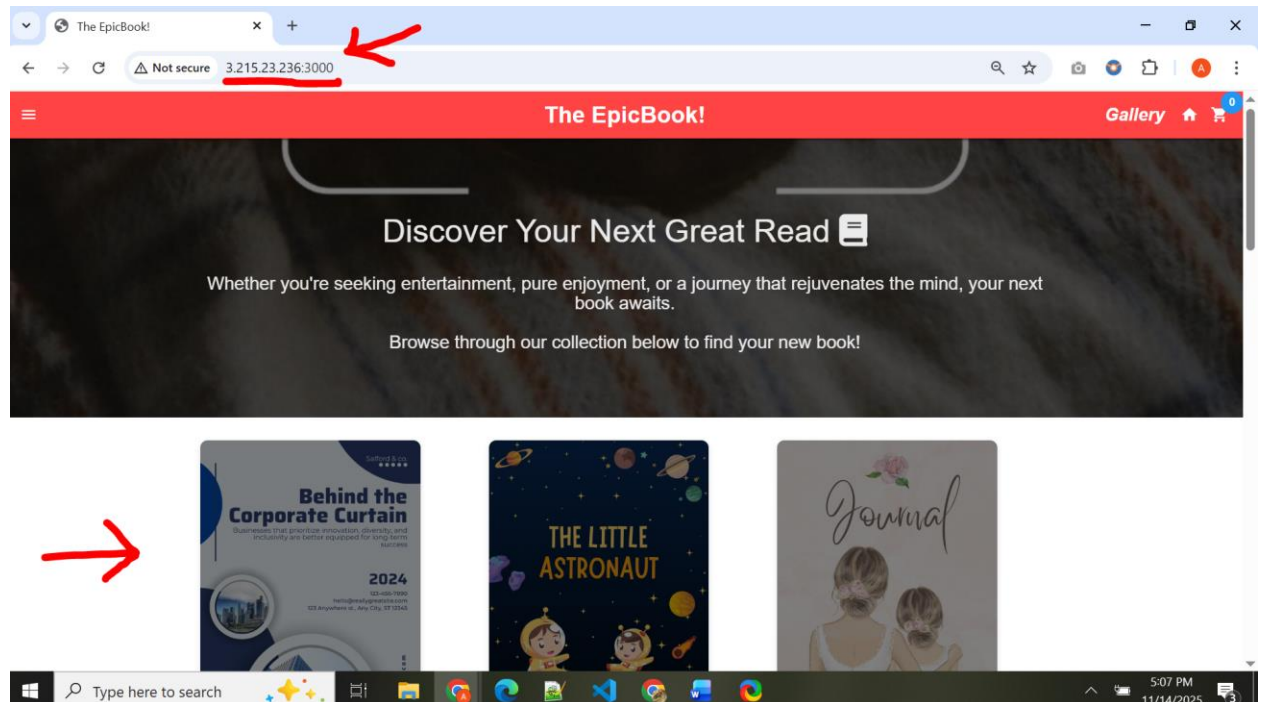
mysql> use bookstore;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> select * from Author;
+-----+
| id | firstName | lastName | createdAt | updatedAt |
+-----+
| 1 | Elin | Hildebrand | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 2 | Brit | Bennett | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 3 | John | Bolton | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 4 | Robin | DiAngelo | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 5 | Delia | Owens | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 6 | John | Grisham | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 7 | Ibrahim X. | Kendi | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 8 | Collette | Ng | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 9 | Ijeoma | Oluo | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 10 | Harper | Lee | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 11 | L.M. | Montgomery | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 12 | Jack | London | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 13 | Jane | Austen | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 14 | F. Scott | Fitzgerald | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 15 | Emily | Bronte | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 16 | Oscar | Wilde | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 17 | George | Orwell | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 18 | J.D. | Salinger | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 19 | Shel | Silverstein | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 20 | Maurice | Sendak | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 21 | Dr. | Suess | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 22 | Beatrix | Potter | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 23 | C.S. | Lewis | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 24 | E.B. | White | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 25 | Eric | Carle | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 26 | Michael | Ende | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 27 | Howard | Zinn | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 28 | Norton | Juster | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 29 | Kurt | Vonnegut | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 30 | Chinua | Achebe | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 31 | Kazuo | Ishiguro | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 32 | Art | Spiegelman | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 33 | Margaret | Atwood | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 34 | Douglas | Adams | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 35 | David Foster | Wallace | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 36 | Philip D. | Morgan | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 37 | Hortense | Spillers | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 38 | Stephen | Best | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 39 | Anthony B. | Pinn | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 40 | Harvey | Young | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 41 | Michael A. | Gomez | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 42 | Sarah | Haley | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 43 | Daryl Michael | Scott | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 44 | Cheryl | Harris | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 45 | Frank | Herbert | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 46 | Andrzej | Sapkowski | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 47 | J.R.R. | Tolkien | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 48 | J.K. | Rowling | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 49 | Stephanie | Meyer | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 50 | George R. R. | Martin | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 51 | Diana | Gabaldon | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 52 | Robert | Jordan | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
| 53 | Cixin | Liu | 2025-11-14 11:31:02 | 2025-11-14 11:31:02 |
+-----+
53 rows in set (0.00 sec)
```

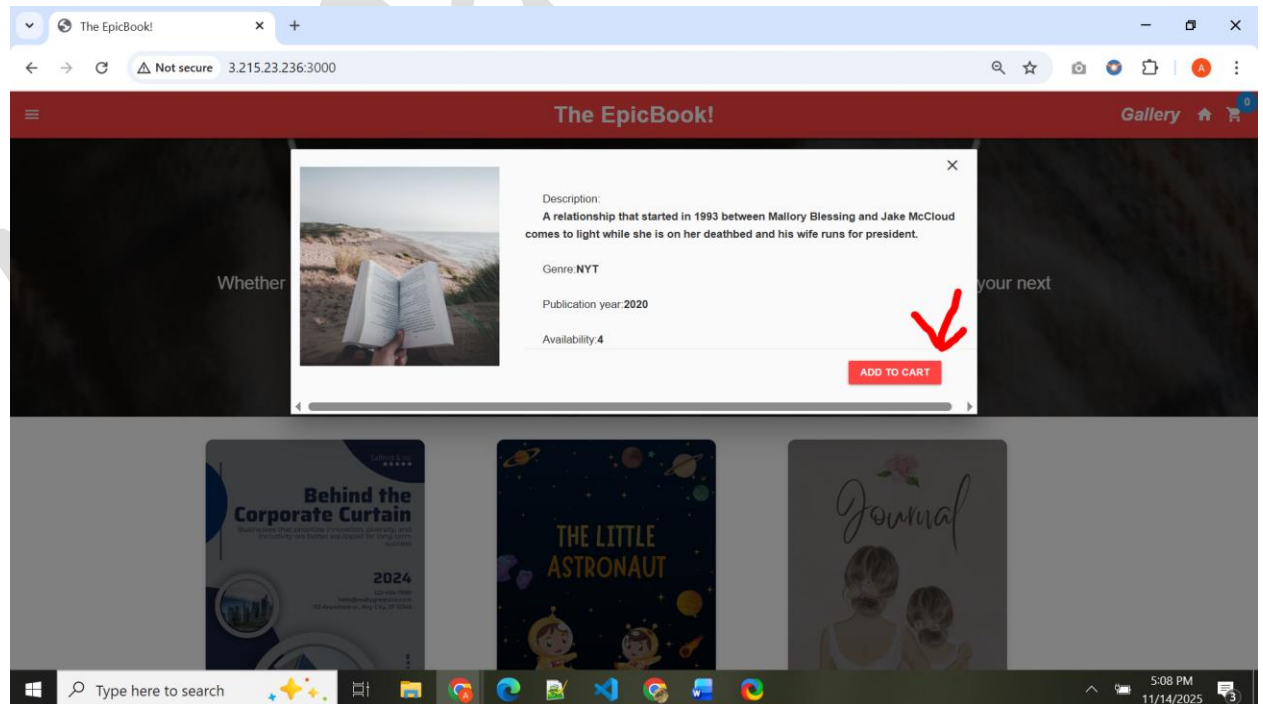

9. Test in Browser

Open your browser:

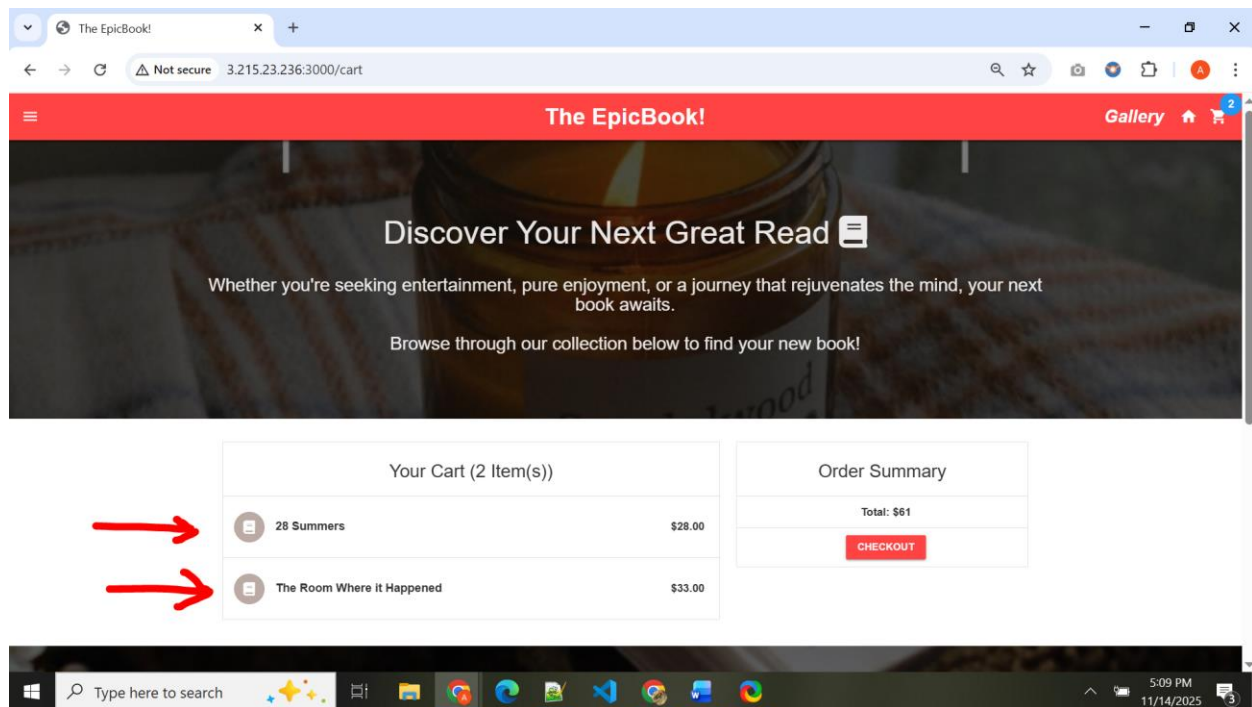
👉 http://<VM_PUBLIC_IP>:3000



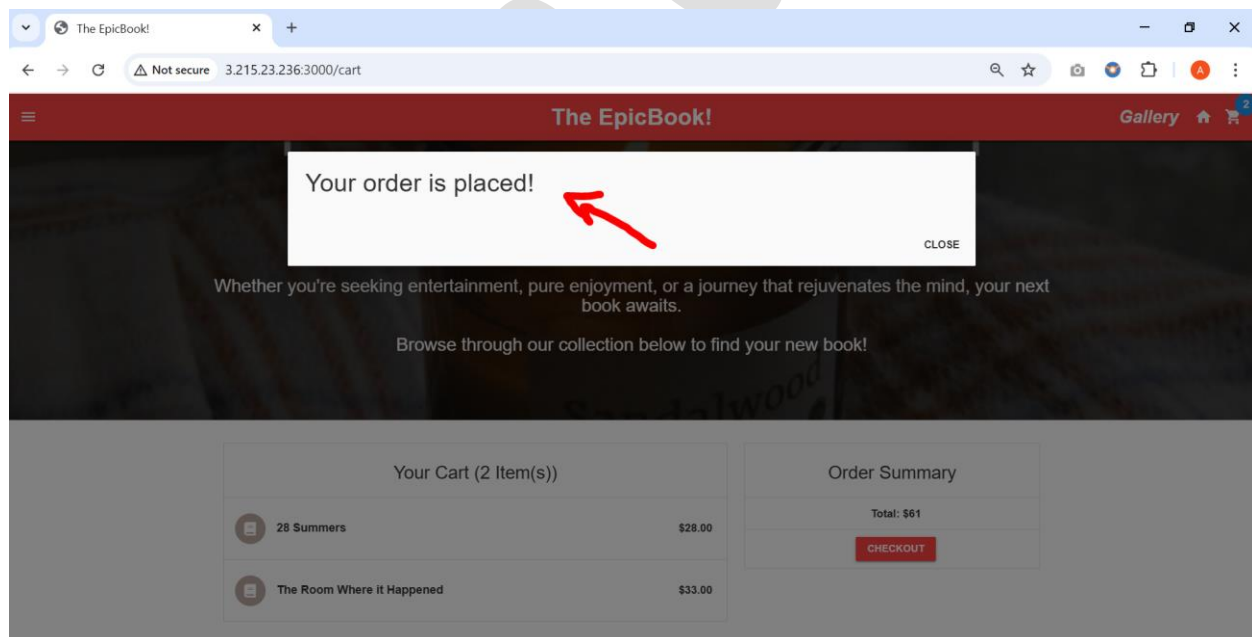
Add Books to the Cart



Go to cart and check if the added books are correctly displaying.



Checkout and "Your order is placed!" message should appear.



Video of Output:



Assignment
49_record_2025-11-1

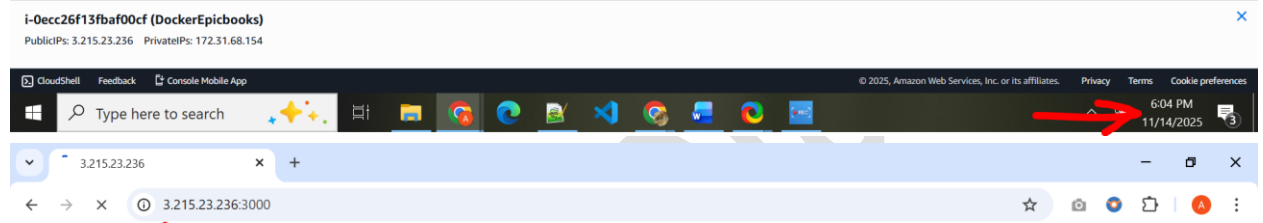
10. Verify Persistence

Run:

`docker-compose down`

(do not add `-v`)

```
ubuntu@ip-172-31-68-154:~/theepicbook$ docker-compose down
Stopping theepicbook_container ... done
Stopping mysql_container ... done
Removing theepicbook_container ... done
Removing mysql_container ... done
Removing network theepicbook_default
ubuntu@ip-172-31-68-154:~/theepicbook$
```



This site can't be reached

3.215.23.236 took too long to respond.

Try:

- Checking the connection
- [Checking the proxy and the firewall](#)
- [Running Windows Network Diagnostics](#)

ERR_CONNECTION_TIMED_OUT

Reload

Details



Then:

`docker-compose up -d`

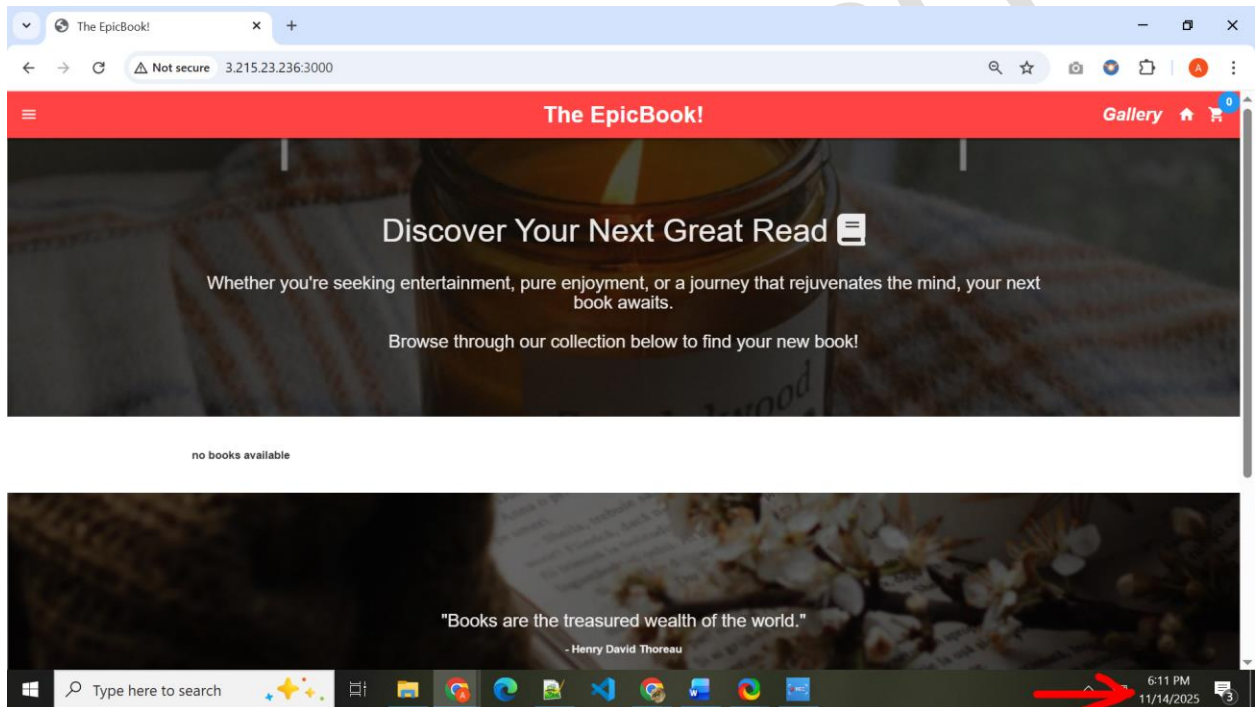
```
ubuntu@ip-172-31-68-154:~/theepicbook$ docker-compose up -d
Creating network "theepicbook_default" with the default driver
Creating mysql container ... done
Creating theepicbook container ... done
ubuntu@ip-172-31-68-154:~/theepicbook$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
68f9d8f91a9f	theepicbook_theepicbook	"docker-entrypoint.s..."	24 seconds ago	Up 23 seconds	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	theepicbook_container
4d501075b4bb	mysql:8.0	"docker-entrypoint.s..."	34 seconds ago	Up 34 seconds (healthy)	0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp	mysql_container

i-0ecc26f13fbaf00cf (DockerEpicbooks)
PublicIPs: 3.215.23.236 PrivateIPs: 172.31.68.154

CloudShell Feedback Console Mobile App © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
Type here to search 6:07 PM 11/14/2025

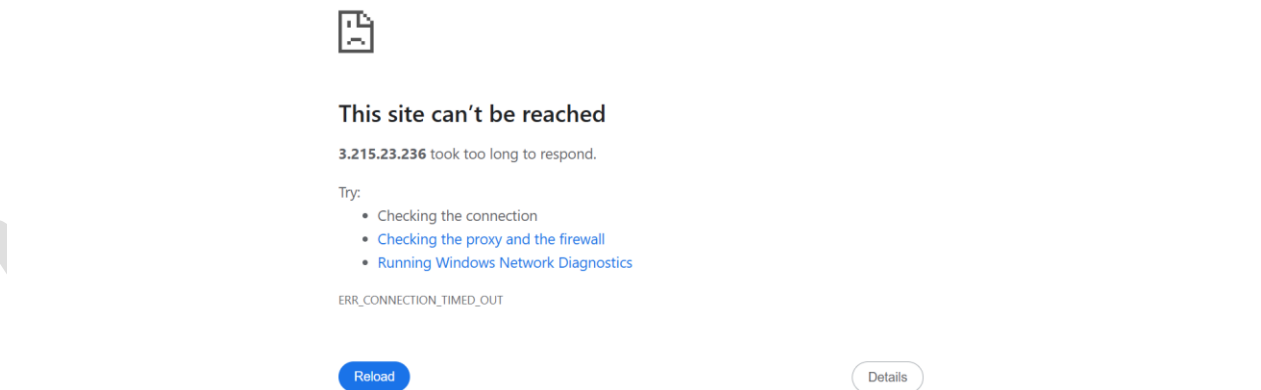
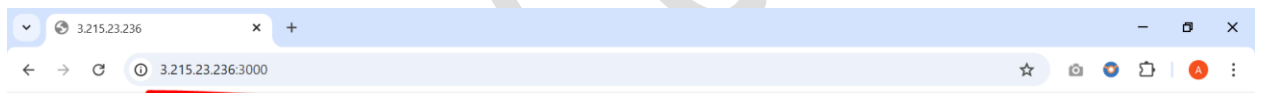
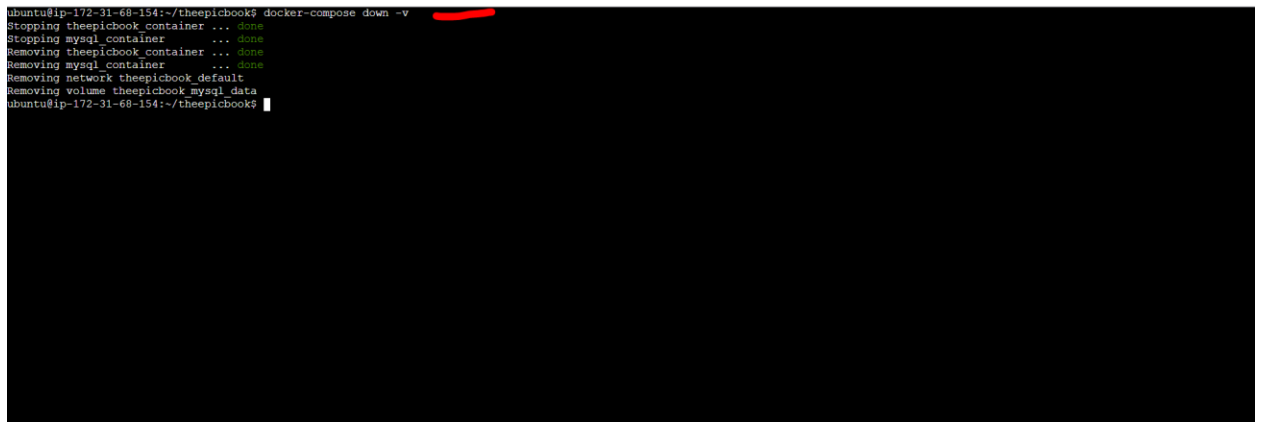
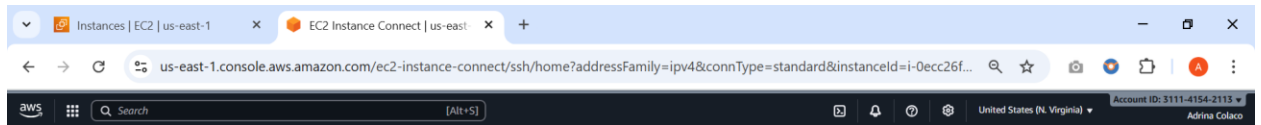
Go back to the app —It should still exist ☒ (stored in mysql_data volume).



11. Teardown (Optional)

To remove everything (including volume):

docker-compose down -v



LinkedIn Post1:

https://www.linkedin.com/posts/adrina-colaco_assignment-43-cloud-vm-bootstrap-docker-activity-7394243277350371328-Rivr?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

LinkedIn post2:

https://www.linkedin.com/posts/adrina-colaco_assignment-44-multi-stage-docker-build-activity-7394245677268197376-ZdQ8?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

LinkedIn post3:

https://www.linkedin.com/posts/adrina-colaco_assignment-45-docker-networking-frontend-activity-7394250349957599233-pECs?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

LinkedIn post4:

https://www.linkedin.com/posts/adrina-colaco_assignment-46-docker-data-persistence-activity-7394260512781381632-kV8G?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

LinkedIn post5:

https://www.linkedin.com/posts/adrina-colaco_assignment-47-docker-compose-one-command-activity-7394308230233001984-nvFp?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

LinkedIn post6:

https://www.linkedin.com/posts/adrina-colaco_assignment-48-deploying-the-book-review-activity-7394610321732083712-GaxA?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhXLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM

LinkedIn post7:

https://www.linkedin.com/posts/adrina-colaco_assignment-49-the-epicbook-capstone-activity-7395081524339105792-

[jsW6?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhxLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM](https://www.linkedin.com/pulse/my-docker-journey-assignments-4349-adrina-colaco-znipcjsW6?utm_source=share&utm_medium=member_desktop&rcm=ACoAAAPhxLwBOfaV_xhbVwfcuz8dVOvN7T9q0RM)

Blog :

<https://www.linkedin.com/pulse/my-docker-journey-assignments-4349-adrina-colaco-znipc>

The End

P.S. This document is part of the **FREE DevOps Micro Internship (DMI) Cohort** run by [Pravin Mishra](#). You may join the next FREE COHORT by filling this [Pre-registration](#) Form.